The Sensorium Manifold:

Emergent Modes of Intelligence

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March 15, 2026

Abstract

Current generative architectures treat sequence modeling as an initial value problem: gradient-trained weights predict the next token before any future constraint can be applied. We reject this framing entirely and cast generation as a *boundary value problem* (BVP). The observer clamps a subset of symbols as fixed constraints and the remaining degrees of freedom are resolved simultaneously via massively parallel geometric resonance search.

We instantiate this paradigm in a physics-native substrate built on three layers:

1. A **Fermat Cube** geometry where each byte value self-addresses a unique face of a $6 \times 4 \times 257$ three-dimensional chord lattice, with sequential structure encoded as composable affine transformations over the Galois field GF(257).
2. A **257-bit Chord** representation where each byte value maps to a deterministic prime-basis bitset, and bitwise operations (`OR`, `AND`, `popcount`) execute discrete wave interference at single-cycle throughput.
3. An **Analytical Phase** navigation layer that maps chords onto a $S^1 \times S^1$ phase torus via their intrinsic prime-index geometry, enabling deterministic closure detection and intent steering without trained co-occurrence models.

No loss function is minimized. No weights are updated. Inference is a single-pass GF(257) geometric distance scan across the entire memory substrate, resolving the nearest rotational state via integer arithmetic on GPU. A two-phase hybrid retrieval pipeline (bitwise filter followed by complex-valued PhaseDial scoring) provides precise candidate ranking. We validate the architecture across text classification, code generation, logical reasoning, phase-dial generalization, image reconstruction, and cross-modal tasks, reporting both successes and failure regimes with equal detail.

0.1 Motivation

The current generative AI stack is structurally constrained by its formulation as an *initial value problem*. Whether through autoregressive transformers or state-space models, generation is treated as strictly sequential extrapolation: given a prefix $x_{0:t}$, compute x_{t+1}, append it, and repeat. This framing imposes three fundamental limitations:

1. **Serial depth bottleneck.** Each step depends on the materialization of all preceding outputs, forcing an $O(N)$ critical-path depth that hardware parallelism cannot bypass.
2. **Irrevocable commitment.** Tokens cannot be retracted once emitted; local errors propagate forward with no mechanism for future constraints to resolve past ambiguity.
3. **Static memory.** Knowledge is compressed into static weight matrices requiring gradient descent to update, causing catastrophic forgetting and forcing reliance on retrieval augmentation.

This paper proposes a departure from the IVP framework. We cast generation as a *boundary value problem*: the observer clamps constraints, and the remaining degrees of freedom are resolved simultaneously by a massively parallel geometric resonance engine operating over a Galois field GF(257).

0.2 Why Bitwise Math Is Wave Interference

The key insight underlying this architecture is that binarized prime-frequency vectors are not merely compressed representations—they are discrete Fourier transforms of semantic waves. Each bit in a 257-bit Chord represents the presence of a specific fundamental oscillator. Bitwise operations then execute actual discrete wave interference:

- `OR` **(Superposition):** Combining tokens in a context window $(A \mid B)$ superposes their oscillator spectra.
- `popcount`$(A \,\&\, B)$ **(Constructive Interference):** Measures the amplitude of shared oscillators between context and memory.
- `popcount`$(A \,\&\, {\sim}B)$ **(Destructive Interference / Noise):** Measures oscillators present in memory but absent from context—harmonic noise or phase mismatch.

Mathematically, the dot product of two binarized vectors equals the popcount of their bitwise AND. The architecture does not abandon wave mechanics; it compiles them down to single-cycle integer operations.

0.3 Paper Organization

The remainder of this paper is organized as follows. Section 0.14 defines the 257-bit Chord primitive and its algebraic operations. Section 0.25 introduces the Fermat Cube geometry and the Icosahedral A_5 extension. Section 0.35 describes the Universal Tokenizer and topological boundary detection. Section 0.19 presents the analytical toroidal phase navigation model. Section 0.4 details the GF(257) resolver pipeline. **??** formalizes generation as a boundary value problem. Section 2 describes topological entropy routing and virtual mitosis. Section 1.1 introduces the dual Substrate/Cortex architecture. Experimental results span text classification, code generation, logical reasoning (bAbI), phase-dial robustness, image reconstruction, cross-domain completion, scaling analysis, and Cortex channel routing.

0.4 Problem Statement

Given a context GF(257) rotation $\mathbf{q} = (a_q, b_q)$ and a graph of N nodes each carrying their own rotation state $\{(a_i, b_i)\}_{i=1}^{N}$, the resolver returns:

$$i^* = \underset{1 \le i \le N}{\arg\min}\ d^2(\mathbf{q}, \mathbf{n}_i) \tag{1}$$

where d^2 is the squared Euclidean distance in the Galois field coordinate space. This is the system's sole inference primitive—all generation, retrieval, and reasoning reduce to repeated resolver calls.

0.5 GF(257) Geometric Distance

Each node in the thermodynamic field carries a composable affine state $(a, b) \in \mathrm{GF}(257)^2$ where:

$$f(x) = (a \cdot x + b) \bmod 257, \qquad a \in \{1, \ldots, 256\},\ b \in \{0, \ldots, 256\} \tag{2}$$

The resolver computes the distance between a query context and every candidate node as:

$$d^2(\mathbf{q}, \mathbf{n}_i) = (a_i - a_q)^2 + (b_i - b_q)^2 \tag{3}$$

This is pure integer arithmetic—no floating-point operations, no popcount, no matrix multiplies. The geometric interpretation is direct: a encodes the *multiplicative* (scaling/rotation) identity, b encodes the *additive* (translational) offset. Nearby nodes in GF(257) space share similar rotational histories.

0.6 Single-Pass Parallel Scan

The resolver executes as a single-pass parallel scan across all N nodes:

1. Each GPU thread evaluates $d^2(\mathbf{q}, \mathbf{n}_i)$ for its assigned node (2 subtractions, 2 multiplications, 1 addition).
2. The distance is inverted to enable lock-free maximum selection: $\hat{s}_i = 131{,}072 - d_i^2$.
3. The score and index are packed into a single `uint64`:

$$\text{packed}_i = \big(\hat{s}_i \ll 32\big) \mid i \tag{4}$$

The winner is determined by a single `AtomicMax` operation per thread. Because the inverted distance occupies the high 32 bits, lexicographic integer comparison correctly preserves distance ordering.

Computational Cost. Each thread executes exactly 5 integer operations (2 subtracts, 2 multiplies, 1 add) plus one atomic compare-and-swap. The entire scan is memory-bound at 4 bytes per candidate (one `GFRotation` = 2 × `uint16`).

0.7 Multi-Backend Dispatch

The kernel dispatch layer supports Metal (Apple Silicon), CUDA (NVIDIA), and CPU (fallback) backends with automatic selection via the `SIX_BACKEND` environment variable:

$$i^* = \underset{i \in \text{Nodes}}{\arg\min}\ d^2(\mathbf{q}, \mathbf{n}_i) \tag{5}$$

For CUDA, multi-GPU orchestration distributes the node array across devices with per-device base offsets, merging local winners via the packed score representation.

0.8 Composition Instead of Accumulation

A key distinction from popcount-based retrieval (BestFill) is that the GF(257) resolver operates on *composed* rotational states rather than accumulated bit densities. When the Sequencer detects a topological boundary event, it emits a rotation:

- **Density Spike** $\to X_{90}$: $(1, 1)$
- **Phase Inversion** $\to Y_{90}$: $(g, 0)$
- **Density Trough** $\to Z_{90}$: $(g, 1)$
- **Low-Variance Flux** $\to X_{180}$: $(1, 2)$

These compose via the group operation:

$$(a_2, b_2) \circ (a_1, b_1) = \big((a_2 \cdot a_1) \bmod 257,\ \ (a_2 \cdot b_1 + b_2) \bmod 257\big) \tag{6}$$

Because composition is non-commutative, the resolver distinguishes sequences that traverse different rotational paths even if they contain identical content—encoding syntax as geometry.

0.9 The Autoregressive IVP Limitation

Contemporary language models formulate generation as an autoregressive Initial Value Problem (IVP): given a starting context, continuously predict the next token. Formally, given prefix $\mathbf{x}_{0:t}$, the IVP computes $p(x_{t+1} \mid \mathbf{x}_{0:t})$ and samples, yielding a Markov chain with strictly left-to-right information flow. This creates three structural limitations:

1. **Serial depth bottleneck:** Each step depends on the materialization of all preceding outputs, forcing an $O(N)$ critical-path depth that hardware parallelism cannot bypass.

2. **Irrevocable commitment:** Tokens cannot be retracted once emitted; local errors propagate as distributional drift—a single decoding error permanently alters the generation trajectory.
3. **Static memory:** Knowledge is compressed into weight matrices $\mathbf{W}$ requiring gradient descent to update, causing catastrophic forgetting.

0.10 BVP Formulation via Topological Vacuum

We frame generation as a *Boundary Value Problem solved through discrete geometric resonance.* Given boundary constraints encoded as a chord $\mathbf{e}$ (the existing context) and a target distribution $\mathbf{t}$ (either the context itself or an ExpectedReality manifold), the system computes the Topological Vacuum:

$$\mathbf{h} = H(\mathbf{t}, \mathbf{e}) = \mathbf{t} \wedge \neg\, \mathbf{e} \tag{7}$$

This isolates the exact prime oscillators *required but missing* from the context. The active bits in $\mathbf{h}$ represent structural constraints that must be satisfied by the continuation; bits not in $\mathbf{h}$ are already satisfied or irrelevant.

0.11 Resonant Fill: The FillScore

Rather than computing transition probabilities, the inference engine searches the entire PrimeField for the manifold block whose topology most perfectly *fills* the vacuum. Given vacuum $\mathbf{h}$ and candidate chord $\mathbf{d}$:

$$\text{FillScore}(\mathbf{h}, \mathbf{d}) = \frac{\texttt{popcount}(\mathbf{h} \wedge \mathbf{d})}{\texttt{popcount}(\mathbf{h}) \;\cdot\; \big(1 + \texttt{popcount}(\mathbf{d} \wedge \neg\, \mathbf{h})\big)} \tag{8}$$

The three terms have precise physical interpretations:

- **Numerator** $\texttt{popcount}(\mathbf{h} \wedge \mathbf{d})$: The number of vacuum bits filled by the candidate—*constructive interference* between the structural need and the candidate's content.
- **Denominator term 1** $\texttt{popcount}(\mathbf{h})$: The total vacuum size, normalizing the score to $[0, 1]$.
- **Denominator term 2** $1 + \texttt{popcount}(\mathbf{d} \wedge \neg\, \mathbf{h})$: A penalty for *structural noise*—candidate oscillators not demanded by the vacuum. The $+1$ prevents division by zero and ensures that a candidate with zero noise achieves the maximum score.

Properties of FillScore.

1. **Bounded:** FillScore $\in [0, 1]$. Achieved at 1 iff $\mathbf{d}$ fills every vacuum bit and introduces no extra bits ($\mathbf{h} \wedge \mathbf{d} = \mathbf{h}$ and $\mathbf{d} \wedge \neg \mathbf{h} = \mathbf{0}$).
2. **Monotone in fill:** For fixed $\mathbf{h}$, adding a vacuum-aligned bit to $\mathbf{d}$ strictly increases the score.
3. **Penalizing noise:** For fixed fill, adding a non-vacuum bit to $\mathbf{d}$ strictly decreases the score via the denominator penalty.
4. $O(D/w)$ **computation:** All three popcount terms are computed in a single pass over the $D/64 = 8$ blocks.

0.12 Transitive Resonance: Analogical Transfer

The chord algebra supports structural analogy without neural weights. Given three facts encoded as chords F_1, F_2, F_3, the system extracts shared factors and forges a hypothesis:

$$B = F_1 \wedge F_2 \qquad \text{(Shared context: “wormhole”)} \tag{9}$$
$$C = F_2 \wedge F_3 \qquad \text{(Shared subject)} \tag{10}$$
$$A = F_1 \wedge \neg B \qquad \text{(Unique to } F_1\text{)} \tag{11}$$
$$D = F_3 \wedge \neg C \qquad \text{(Unique to } F_3\text{)} \tag{12}$$
$$H = A \vee D \qquad \text{(Forged hypothesis)} \tag{13}$$

This implements the analogy $A : B :: C : D$ using five bitwise operations in $O(D/w)$ time, enabling zero-shot relational transfer.

0.13 The Cantilever Generation Loop

Generation extends the original prompt base like a physical cantilever:

1. Build the context manifold $\mathbf{e}_0$ from the prompt chords.
2. Compute the structural vacuum $\mathbf{h}_t = H(\mathbf{t}, \mathbf{e}_t)$.
3. Execute BestFill across all 5 Fibonacci planes simultaneously, returning $\hat{\mathbf{d}} = \arg\max_{\mathbf{d} \in \mathrm{PF}} \mathrm{FillScore}(\mathbf{h}_t, \mathbf{d})$.
4. Check EigenMode intent alignment: $\Delta\theta(\hat{\mathbf{d}}) < \delta_{\text{intent}}$.
5. Integrate: $\mathbf{e}_{t+1} = \mathbf{e}_t \vee \hat{\mathbf{d}}$.
6. Emit the byte-level readout of $\hat{\mathbf{d}}$.
7. Repeat until geometric closure is detected (Section 0.21) or $t > t_{\max}$.

The growing structure becomes its own foundation for subsequent searches, forming a positive feedback loop where each successful fill sharpens the context for the next.

0.14 Deterministic Prime-Basis Identity

Let $D = 257$ denote the chord dimensionality and let $\mathcal{B} = \{0, \ldots, 255\}$ be the byte alphabet. Every byte value $b \in \mathcal{B}$ is mapped to a deterministic binary vector $\mathbf{c}(b) \in \{0,1\}^D$ by activating exactly five bits at positions given by coprime spreading:

$$\mathcal{P}(b) = \big\{7b \bmod D,\ 13b \bmod D,\ 31b \bmod D,\ 61b \bmod D,\ 127b \bmod D\big\} \tag{14}$$

$$c_i(b) = \begin{cases} 1 & \text{if } i \in \mathcal{P}(b), \\ 0 & \text{otherwise.} \end{cases} \tag{15}$$

Because the multipliers $\{7, 13, 31, 61, 127\}$ are pairwise coprime and $D = 257$ is prime, distinct byte values receive distinct activation patterns (verified by exhaustive enumeration over all 256 values). The resulting chord is extremely sparse: $\|\mathbf{c}(b)\|_1 = 5$ out of $D = 257$ bits, yielding a sparsity ratio of $5/257 \approx 1.95\%$.

Why Coprime Multipliers. The choice of coprime multipliers is not arbitrary. Multiplying by a value p and reducing modulo a prime $D = 257$ generates a permutation of the residues $\{0, \ldots, D-1\}$ for any $p \not\equiv 0 \pmod{D}$. Since all five multipliers are themselves prime and less than 257, each independently generates a distinct full-period permutation, maximizing the minimum Hamming distance between any two byte chords.

0.15 Algebraic Operations

The Chord supports a closed algebra of compositional operations. Each executes in $O(D/w)$ time where w is the SIMD word width (64 for `uint64`, yielding $D/w = 8$ operations per chord):

$$\textbf{ChordOR:} \quad (\mathbf{a} \vee \mathbf{b})_i = a_i \vee b_i \qquad \text{(Superposition / LCM)} \tag{16}$$

$$\textbf{ChordGCD:} \quad (\mathbf{a} \wedge \mathbf{b})_i = a_i \wedge b_i \qquad \text{(Shared factors / Intersection)} \tag{17}$$

$$\textbf{ChordHole:} \quad H(\mathbf{t}, \mathbf{e})_i = t_i \wedge \neg e_i \qquad \text{(Structural Vacuum)} \tag{18}$$

$$\textbf{Similarity:} \quad \mathrm{sim}(\mathbf{a}, \mathbf{b}) = \sum_{i=0}^{D-1} a_i \cdot b_i = \texttt{popcount}(\mathbf{a} \wedge \mathbf{b}) \qquad \text{(Discrete dot product)} \tag{19}$$

Equivalence to Inner Product. For binary vectors $\mathbf{a}, \mathbf{b} \in \{0,1\}^D$, the real-valued inner product $\langle \mathbf{a}, \mathbf{b} \rangle$ reduces exactly to the popcount of their bitwise AND:

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=0}^{D-1} a_i \, b_i = \big|\{i : a_i = 1 \wedge b_i = 1\}\big| = \texttt{popcount}(\mathbf{a} \,\&\, \mathbf{b}) \tag{20}$$

This identity is the mechanism by which single-cycle integer instructions execute semantic similarity computation.

0.16 Positional Binding via Circular Shift

To encode sequential position without destroying content, we define the *RollLeft* operator as a circular permutation over bit indices:

$$\mathrm{RollLeft}(\mathbf{c},\, s)_i = c_{(i-s) \bmod D} \tag{21}$$

Because circular shifts are invertible and non-commutative with superposition, positional binding preserves all algebraic properties while ensuring that re-ordered sequences produce distinct chords:

$$\mathrm{RollLeft}(\mathbf{c}_d, 0) \vee \mathrm{RollLeft}(\mathbf{c}_o, 1) \vee \mathrm{RollLeft}(\mathbf{c}_g, 2) \;\neq\; \mathrm{RollLeft}(\mathbf{c}_g, 0) \vee \mathrm{RollLeft}(\mathbf{c}_o, 1) \vee \mathrm{RollLeft}(\mathbf{c}_d, 2) \tag{22}$$

That is, ``dog'' $\neq$ ``god'' in the chord representation.

0.17 SimHash Structural Binning

The *ChordBin* operator maps any chord $\mathbf{c}$ to an 8-bit structural bin $b \in \{0, \ldots, 255\}$ via a SimHash-style random hyperplane projection. Given $k = 8$ fixed seed values $\{s_1, \ldots, s_8\}$, the accumulator for plane j sums signed contributions from each active bit:

$$\mathrm{acc}_j = \sum_{i : c_i = 1} \begin{cases} +1 & \text{if } h(i, s_j) \geq 2^{63}, \\ -1 & \text{otherwise,} \end{cases} \qquad h(i, s_j) = i \cdot s_j + (i \ll (j+1)) \tag{23}$$

The bin is then the 8-bit integer whose j-th bit is $[\mathrm{acc}_j \geq 0]$. Chords with high Hamming similarity map to the same or nearby bins with high probability, enabling the EigenMode phase table (Section 0.19) to use a fixed-size 256×256 co-occurrence matrix without per-chord storage.

0.18 FlatChord: GPU-Optimized Dense Representation

For GPU dispatch, the sparse 257-bit Chord is converted to a *FlatChord*: a dense array of at most D active prime indices $\{p_1, \ldots, p_k\}$ where $k = \|\mathbf{c}\|_1$. This eliminates bit-twiddling thread divergence in SIMT architectures: every thread iterates over exactly k entries rather than scanning all $D/64 = 8$ blocks.

Batched flattening across N chords is parallelized via a pre-warmed, auto-scaling worker pool, yielding amortized $O(k)$ per chord with near-zero goroutine creation overhead.

0.19 Analytical Phase Model

Unlike spectral methods that require building co-occurrence matrices and extracting eigenvectors, the phase model derives toroidal coordinates *analytically* from the intrinsic bit distribution of each chord over GF(257). No training data is required; every chord's position on the phase torus is determined at construction time.

0.19.1 Phase Assignment via Prime Geometry

Each chord $\mathbf{c}$ has $k = \|\mathbf{c}\|_1$ active bits at positions $\{p_1, \ldots, p_k\} \subset \{0, \ldots, 256\}$. We assign toroidal coordinates $(\theta, \phi) \in [-\pi, \pi)^2$ directly:

Theta (Circular Mean of Active Positions). The angular identity of a chord is the circular mean of its active prime indices, uniformly distributed around $[0, 2\pi)$ over the 257-element field:

$$\theta(\mathbf{c}) = \operatorname{atan2}\Big(\sum_{j=1}^{k} \sin\big(\tfrac{2\pi\, p_j}{257}\big),\ \sum_{j=1}^{k} \cos\big(\tfrac{2\pi\, p_j}{257}\big)\Big) \tag{24}$$

This phase is geometrically equivalent to a RotateX translation on the Fermat Cube: chords with similar prime activation patterns occupy nearby θ-positions on the torus.

Phi (Density Phase). The secondary phase encodes the chord's population density:

$$\phi(\mathbf{c}) = \frac{2\pi \cdot \|\mathbf{c}\|_1}{257} \tag{25}$$

Sparse chords (low popcount) sit near $\phi = 0$; dense chords approach $\phi = 2\pi$. The pair (θ, ϕ) embeds every chord onto the 2-torus $\mathbb{T}^2 = S^1 \times S^1$.

0.20 Sequential Phase Tracking and Intent Alignment

During generation, the accumulated sequential phase is tracked as a *weighted circular mean* over the emitted chord sequence $(\mathbf{c}_1, \ldots, \mathbf{c}_n)$:

$$\bar{\theta}_n = \operatorname{atan2}\Big(\sum_{i=1}^{n} w_i \sin\theta(\mathbf{c}_i),\ \sum_{i=1}^{n} w_i \cos\theta(\mathbf{c}_i)\Big) \tag{26}$$

where $w_i = \|\mathbf{c}_i\|_1$ weights each chord by its active count (density). The *concentration parameter* κ_n (analogous to the κ parameter of the von Mises distribution) measures phase coherence:

$$\kappa_n = \frac{\sqrt{S_n^2 + C_n^2}}{W_n}, \qquad S_n = \sum_{i=1}^{n} w_i \sin\theta_i, \quad C_n = \sum_{i=1}^{n} w_i \cos\theta_i, \quad W_n = \sum_{i=1}^{n} w_i \tag{27}$$

Values of κ_n near 1 indicate strong phase alignment (the trajectory is coherent); values near 0 indicate phase diffusion (semantic drift).

0.21 Geometric Closure Detection

Generation terminates when the candidate chord sequence's weighted circular mean phase returns to within a configurable threshold δ of the anchor phase $\bar{\theta}_{\text{anchor}}$ established at prompt initialization:

$$\Delta\theta = \min\big(|\bar{\theta}_n - \bar{\theta}_{\text{anchor}}|,\ \ 2\pi - |\bar{\theta}_n - \bar{\theta}_{\text{anchor}}|\big) < \delta \tag{28}$$

where δ is derived from the architecture's Shannon capacity parameter (`shannonCapacity`, default 4.5 radians). This constitutes a topological proof of structural closure: the generative trajectory has completed a semantically coherent loop on the torus. Because $\mathbb{T}^2$ is a compact manifold, every trajectory is pre-compact (bounded), preventing entrapment in isolated fixed points or periodic orbits.

0.22 Advantages of the Analytical Model

- **Zero training cost.** No co-occurrence matrices, no eigendecomposition, no corpus-dependent calibration. The phase map is a pure function of the chord's bit pattern.
- **Deterministic.** Identical chords always map to identical (θ, ϕ) regardless of context, ensuring reproducibility.
- $O(k)$ **per chord.** Phase computation scales with the number of active bits ($k \leq 257$), not with corpus size.
- **Intrinsically geometric.** The prime-index circular mean directly corresponds to the chord's position in the GF(257) rotational lattice, making phase navigation and Galois field operations composable.

0.23 Two-Phase Retrieval Pipeline

The HybridSubstrate provides an alternative retrieval pathway that combines the speed of bitwise filtering with the precision of continuous complex scoring:

1. **Phase 1 — Bitwise Filter:** All memory entries are scored against the context Chord using popcount-based resonance (identical to BestFill Pass 4). The top-K candidates are retained.
2. **Phase 2 — PhaseDial Scoring:** The surviving candidates are evaluated against a complex-valued *PhaseDial* fingerprint using full complex cosine similarity:

$$\text{sim}(\mathbf{a}, \mathbf{b}) = \frac{\text{Re}(\sum_i \bar{a}_i\, b_i)}{\|\mathbf{a}\| \cdot \|\mathbf{b}\|} \tag{29}$$

3. **Phase 3 — Readout:** The payload of the highest-scoring entry is returned.

0.24 Geodesic Phase Scanning

The GeodesicScan sweeps a seed PhaseDial fingerprint through 360° of phase rotation in discrete steps, ranking all candidates at each angle. This produces a full rotational landscape revealing:

- The phase angle at which margin (score gap between 1st and 2nd candidates) is maximized—indicating maximum discriminability.
- Entropy variations across the rotation, revealing structural symmetries in the memory substrate.

0.25 Structural Layout

The base geometric primitive is a *Fermat Cube*—a three-dimensional array of $6 \times 4 \times 257 = 6{,}168$ Chord positions. Each of the 6 physical sides carries 4 rotational orientations (0°, 90°, 180°,

270°), and each orientation holds 257 chords: 256 data faces (one per byte value) plus a control chord at face index 256.

```
type Side [4][CubeFaces]data.Chord   // 4 rotations × 257 chords
type Cube struct {
    Sides [6]Side                     // 6 physical faces
}
```

The control chord at face 256 is structurally protected: data ingestion never writes to it. In the Cortex (§1.1), the control chord determines edge opcodes between connected cubes.

All rotational addressing is performed via composable affine transformations over the Galois field GF(257):

$$f(x) = (a \cdot x + b) \pmod{257}, \qquad a \in \{1, \ldots, 256\},\ b \in \{0, \ldots, 256\} \tag{30}$$

Because $257 = 2^8 + 1$ is a Fermat prime, GF(257)* is cyclic of order 256—every non-zero element generates the full multiplicative group. This ensures that *every* byte value maps bijectively to a unique face, and composition of rotations is closed under the group operation (§0.8).

0.26 The Icosahedral Manifold and A_5 Extension

Each memory primitive is pre-allocated as an `IcosahedralManifold`: 5 Fermat Cubes stored contiguously with a compact orientational header:

```
type IcosahedralManifold struct {
    Header  ManifoldHeader   // 16-bit orientational state
    Cubes   [5]FermatCube    // 5 × 257 = 1,285 chord faces per cube
}
```

In flat addressing mode (one cube), total state per Fermat Cube is $6 \times 4 \times 257 \times 257 = 1{,}585{,}176$ bits.

In baseline Cubic mode (`State=0`), only `Cubes[0]` is active.

A_5 Macro-Permutations. Post-mitosis, macro-topology is governed by the alternating group $A_5 \cong \mathrm{Sym}(\text{Icosahedron})$, which consists exclusively of *even permutations* of 5 elements. Three generator types suffice:

$$\textbf{3-Cycle:} \quad \pi_{(a\,b\,c)} : \mathrm{Cubes}[a] \to \mathrm{Cubes}[b] \to \mathrm{Cubes}[c] \to \mathrm{Cubes}[a] \tag{31}$$

$$\textbf{Double Transposition:} \quad \pi_{(a\,b)(c\,d)} : \mathrm{Cubes}[a] \leftrightarrow \mathrm{Cubes}[b],\ \ \mathrm{Cubes}[c] \leftrightarrow \mathrm{Cubes}[d] \tag{32}$$

$$\textbf{5-Cycle:} \quad \pi_{(a\,b\,c\,d\,e)} : \mathrm{Cubes}[a] \to \mathrm{Cubes}[b] \to \cdots \to \mathrm{Cubes}[e] \to \mathrm{Cubes}[a] \tag{33}$$

Each permutation executes as an array index swap in $O(1)$ without data copying.

0.27 The 16-Bit Rotation Header

A compact `uint16` header tracks the full orientational history:

Bit 15	**Bits 14–9**	**Bits 8–5**	**Bits 4–0**
State {Cubic, Ico}	RotState $\in [0, 59]$	Winding mod 16	Reserved —

The **RotState** indexes the precomputed Unified Geodesic Matrix—a symmetric 60×60 lookup table storing pairwise angular distances $d(r_i, r_j) = \arccos\big(|q_i \cdot q_j|\big)$ between all A_5 orientations, requiring only $60 \times 60 = 3{,}600$ bytes (fits in L1 cache). Because no floating-point quaternion math occurs during encoding, quaternion drift is physically impossible.

0.28 Virtual Mitosis: $O(1)$ Phase Transition

Mitosis triggers when the global macro-density of `Cubes[0]` reaches a precise threshold:

$$\rho_0 = \frac{\text{total active bits in Cubes[0]}}{\text{total chord positions} \times 257} \geq 0.45 \tag{34}$$

The transition is a single-cycle state flip:

1. `Cubes[0]` retains its exact saturated state.
2. `Cubes[1..4]` remain zeroed (empty orthogonal subspaces ready to receive entropy).
3. `State` flips from 0 to 1, unlocking 60 rotational states.

De-Mitosis (Reversible Collapse). When global density across all 5 cubes drops below a sparsity threshold, the Geodesic LUT identifies the minimum-energy A_5 permutation to return `Cubes[0]` to canonical orientation, the dispersed attributes of `Cubes[1..4]` are pooled back via majority-vote fractal condensation, and the State reverts to 0.

Entropy Budget. In Cubic mode ($|\mathbf{O}| = 24$), the rotational register carries $\log_2 24 \approx 4.58$ bits of structural entropy. After mitosis ($|A_5| = 60$), this expands to $\log_2 60 \approx 5.91$ bits. Combined with the 4-bit winding counter ($\log_2 16 = 4$ bits), the total orientational entropy budget is ≈ 9.91 bits per manifold—sufficient to distinguish $2^{9.91} \approx 961$ unique structural configurations.

0.29 Contiguous GPU-Dispatch Array

The PrimeField is the compute-side representation of memory: a flat, contiguous array of `IcosahedralManifold` structures that the GPU scans in parallel. Each manifold occupies a fixed 8.64 KB regardless of fill state, guaranteeing stable pointer arithmetic and aligned warp strides.

- **Insertion:** Tokens are merged into the active manifold via `ChordOR` at a block index determined by the token's topological events and chord dynamics.
- **Rotation:** Structural events physically permute the active manifold's block array. Because A_5 permutations are applied as array index remappings, they execute in $O(1)$ without data copying.
- **Atomic Snapshots:** Read access returns a `(pointer, N)` pair under atomic load, preventing concurrency tearing during background GPU dispatch.

0.30 Morton-Coded Spatial Indexing

Each token is assigned a deterministic 64-bit *Morton key* that packs three coordinates into a single integer for locality-preserving sort order:

$$\mathcal{M}(z, \text{sym}, \text{pos}) = \big(\texttt{uint64}(z) \ll 56\big) \;\big|\; \big(\texttt{uint64}(\text{sym}) \ll 32\big) \;\big|\; \texttt{uint64}(\text{pos}) \tag{35}$$

The bit-field layout from MSB to LSB is:

Bits 63–56	**Bits 55–32**	**Bits 31–0**
z (Scale Index) 8 bits	Symbol (Byte Value) 24 bits	Position (Sequence Offset) 32 bits

This layout is load-bearing in three ways:

1. **Scale stratification:** Grouping by z in the MSBs ensures that all entries at the same hierarchical depth sort contiguously, enabling $O(\log N)$ binary search within a scale layer.
2. **Symbol locality:** Within a scale, all occurrences of the same byte value occupy a contiguous key range $[\text{sym} \ll 32,\ (\text{sym}+1) \ll 32 - 1]$, allowing `QueryByByte` to narrow a 50M-entry store to a single binary-search band.
3. **Positional contiguity:** Within a symbol band, positions are numerically ordered in the LSBs, making neighborhood queries (`QueryNeighborhood`) a simple range scan.

0.31 Zipfian Distributions and Collision-as-Compression

Natural data sources—text, images, audio—universally exhibit *Zipfian* frequency distributions: a small number of byte-position patterns account for the overwhelming majority of occurrences. The LSM Spatial Index exploits this directly.

When two tokens share the same Morton key (identical byte value at the same scale and position), the LSM merge operation natively *coalesces* them into a single entry. This is not a hash collision requiring resolution—it is lossless deduplication:

$$\text{Compression Ratio} = \frac{\text{Raw Token Count}}{\text{Unique Morton Keys}} = \frac{N}{|\{\mathcal{M}(z_i, s_i, p_i)\}_{i=1}^{N}|} \tag{36}$$

Because Zipf's Law guarantees that the frequency of the r-th most common pattern scales as $f(r) \propto r^{-\alpha}$ with $\alpha \approx 1$, the number of unique keys grows sub-linearly with corpus size. Empirically, we observe that compression ratios **increase** as datasets grow larger, because larger corpora sample the tail of the Zipfian distribution more densely, producing proportionally more collisions per novel entry.

Table 1: Collision-compression metrics across datasets. Ratios increase with dataset size, consistent with Zipfian redundancy scaling.

Dataset	**Raw Tokens**	**Unique Keys**	**Dedup. %**	**Compression ×**
CIFAR-10 (60K images)	195,000,000	3,693,182	98.1%	52.8×
MBPP (Python)	—	—	—	—
bAbI (QA)	—	—	—	—
Combined Corpus	—	—	—	—

The theoretical lower bound on compression is the dataset's Shannon entropy H: the system cannot compress below the information content of the source distribution. The gap between observed compression and the entropy bound measures the efficiency of the Morton key design.

0.32 Entropy-Driven Sequence Boundaries

When the Sequencer detects a topological boundary and the active manifold carries signal (non-empty chords), the PrimeField freezes the current manifold and advances to a new one. The Z-coordinate of the Morton key tracks scale depth: each sequence reset increments z, stratifying memory into episodic layers.

0.33 Prototype Snapping and Cleanup

To stabilize the memory substrate against noise, the PrimeField maintains per-block prototype attractors. Incoming chords are compared against stored prototypes; if similarity exceeds a threshold, the chord is "snapped" to the prototype—reinforcing established patterns. Novel

chords that diverge sufficiently seed new prototypes, enabling the substrate to discover and crystallize recurring structural motifs.

0.34 Manifold Masking for Self-Exclusion

During BestFill search, the system can `Mask` individual manifolds (temporarily zeroing them) to exclude recently matched spans from consideration. This prevents the popcount echo-attractor problem where the same high-scoring span is retrieved repeatedly. Masked manifolds are restored via `Unmask` after the search completes.

0.35 The Universal Tokenizer

The Universal Tokenizer converts raw byte streams into geometric tokens without learned vocabularies or modality-specific encoders. Each byte occurrence at position *pos* with scale index z and symbol value s produces one Token:

$$\text{TokenID} = \mathcal{M}(z, s, pos) \tag{37}$$

where $\mathcal{M}$ is the Morton code (Equation (35)). The corresponding Chord is the deterministic $\mathbf{c}(s)$ from Equation (14), bound to its sequential position via RollLeft (Equation (21)).

From the moment data is tokenized, the system never examines raw byte values again until final output rendering. All downstream computation operates exclusively on Chords and TokenIDs.

0.36 The Topological Sequencer

The Sequencer segments incoming data streams into sequences based on *topological properties* of the data itself, not on corpus-derived statistics (BPE) or fixed window sizes.

Step 1: Measurement. For each token at position t with chord $\mathbf{c}_t$, the Sequencer extracts two geometric signals:

$$\text{pop}_t = \|\mathbf{c}_t\|_1 = \texttt{popcount}(\mathbf{c}_t) \qquad \text{(Geometric population count)} \tag{38}$$

$$\text{phase}_t = \sqrt{\Theta_{b_t}^2 + \Phi_{b_t}^2} \qquad \text{(Toroidal phase magnitude)} \tag{39}$$

where $b_t = \text{ChordBin}(\mathbf{c}_t)$ and Θ, Φ are the EigenMode phase tables (**??**).

Step 2: Golden Ratio Exponential Smoothing. Running estimates are maintained via exponential moving averages with the golden ratio as the decay constant:

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad \alpha = \varphi^{-3} \approx 0.236 \tag{40}$$

$$\overline{\text{pop}}_t = (1-\alpha)\,\overline{\text{pop}}_{t-1} + \alpha \cdot \text{pop}_t \tag{41}$$

$$\overline{\text{phase}}_t = (1-\alpha)\,\overline{\text{phase}}_{t-1} + \alpha \cdot \text{phase}_t \tag{42}$$

The golden ratio is not arbitrary: φ^{-3} provides a smoothing constant that balances responsiveness to structural changes against noise suppression, with the irrational decay rate preventing aliasing with any periodic structure in the data.

Step 3: Welford's Online Variance. The absolute deviations $\Delta_{\text{pop}} = |\text{pop}_t - \overline{\text{pop}}_t|$ and $\Delta_{\text{phase}} = |\text{phase}_t - \overline{\text{phase}}_t|$ are tracked using Welford's single-pass online variance algorithm, which maintains running mean $\bar{x}_n$ and sum of squared deviations $M_{2,n}$:

$$\delta_n = \Delta_n - \bar{x}_{n-1} \tag{43}$$

$$\bar{x}_n = \bar{x}_{n-1} + \frac{\delta_n}{n} \tag{44}$$

$$M_{2,n} = M_{2,n-1} + \delta_n \cdot (\Delta_n - \bar{x}_n) \tag{45}$$

The unbiased sample standard deviation is $\hat{\sigma}_n = \sqrt{M_{2,n}/(n-1)}$.

Step 4: Dynamic Z-Score Thresholding. Boundary events are triggered when absolute deviations exceed adaptive thresholds derived from the running variance and shared calibrator sensitivity targets $\gamma_{\text{pop}}, \gamma_{\text{phase}}$:

$$\tau_{\text{pop}} = \bar{\Delta}_{\text{pop}} + \hat{\sigma}_{\text{pop}} \cdot \gamma_{\text{pop}} \tag{46}$$

$$\tau_{\text{phase}} = \bar{\Delta}_{\text{phase}} + \hat{\sigma}_{\text{phase}} \cdot \gamma_{\text{phase}} \tag{47}$$

An additional hard topological boundary fires when:

$$\Delta_{\text{phase}} > \frac{\pi}{\kappa} \tag{48}$$

where κ is the `FrequencySpread` hyperparameter.

Step 5: Event Classification. Boundary events are classified into topological triggers that drive the non-commutative rotations of the manifold (Section 0.25):

Event	Condition	Manifold Action
DensitySpike	$\Delta_{\text{pop}} > \tau_{\text{pop}}$ and $\text{pop}_t > \overline{\text{pop}}_t$	3-Cycle $(0\,1\,2)$
DensityTrough	$\Delta_{\text{pop}} > \tau_{\text{pop}}$ and $\text{pop}_t < \overline{\text{pop}}_t$	3-Cycle $(0\,2\,1)$
PhaseInversion	$\Delta_{\text{phase}} > \tau_{\text{phase}}$ or $\Delta_{\text{phase}} > \pi/\kappa$	Double transposition $(0\,3)(1\,4)$
LowVarianceFlux	coherence time > 10	5-Cycle $(0\,1\,2\,3\,4)$

0.37 Closed-Loop Calibration

The Sequencer incorporates bidirectional feedback at two timescales:

- **High-frequency (local):** When a boundary fires, the local chord density $\rho = \|\mathbf{c}\|_1/D$ is compared against target density bounds $[\rho_{\min}, \rho_{\max}]$. If $\rho > \rho_{\max}$, the sensitivity γ is decreased by $\varphi^{-9} \approx 0.013$; if $\rho < \rho_{\min}$, it is increased symmetrically.
- **Low-frequency (global):** BestFill retrieval quality reports adjust calibration at rate $\varphi^{-5} \approx 0.090$: over-discrimination (missed relevant matches) raises γ to widen boundaries; under-discrimination (matched unrelated content) lowers γ to sharpen boundaries.

Local-to-global synchronization uses a median smoothing merge at rate $\varphi^{-6} \approx 0.055$, ensuring that fast local adaptations propagate gradually to the global state without oscillatory instability.

0.38 Machine Architecture

The Machine is the top-level entrypoint to the architecture. It orchestrates data loading, PrimeField construction, and the prompt-to-generation loop:

1. **Loading:** Raw byte streams are tokenized via the Universal Tokenizer, sequenced by the topological Sequencer, and inserted into the PrimeField with their structural events driving manifold rotations.
2. **EigenMode Calibration:** After ingestion, a multi-scale co-occurrence matrix is built from the stored chords and factored into toroidal eigenvectors—establishing the navigation phase map.
3. **Prompt Processing:** Input prompt bytes are converted to chords, accumulated into a query manifold via topological event application, and used to seed the generation context.

0.39 The Generation Loop

Generation proceeds as follows (up to `maxGenerationSteps = 256`):

1. Derive a *slotMask* by comparing the query context against the ExpectedReality manifold, identifying which blocks contain observed data and which represent structural vacuums.
2. Execute BestFill across the PrimeField with multi-hop reasoning: up to `maxReasoningHops = 3` iterative searches, each time masking the previously matched manifold to encourage exploration.
3. Integrate the matched span's chords into the context manifold via the slotMask, only filling blocks that were identified as missing.
4. Decode the best-matching manifold's readout bytes and emit them.
5. Check for geometric closure via EigenMode phase convergence, or detect manifold collapse (score below threshold).

0.40 Multi-Hop Reasoning via Iterative Masking

For reasoning tasks requiring information from multiple memory locations, the Machine supports multi-hop retrieval. After each BestFill match, the matched manifold is temporarily masked (Section 0.34) and the search is repeated. This forces the system to recruit increasingly distant structural neighbors, enabling transitive inference chains.

1 Architecture

1.1 The Dual Architecture: Substrate and Cortex

The architecture introduces a fundamental bifurcation between persistent memory and volatile reasoning, formalized as two structural layers operating over the same underlying coordinate geometry of the Fermat Cube.

1.1.1 The Substrate: Persistent Holographic Storage ($Z = 0$)

The Substrate functions as the long-term memory of the system. It eschews traditional continuous embeddings in favor of a **Thermodynamic Trie**, where incoming tokens are routed deterministically based on their intrinsic geometric identity (`IntrinsicFace`). Specifically, every byte addresses a unique face of the Fermat Cube. This self-addressing mechanism completely eliminates hash collisions and projection matrices.

Chronological generation in the Substrate is treated deterministically as an affine operation over the Galois field GF(257). Rotations induced by sequential input accumulate as affine transforms $f(x) = (ax + b) \pmod{257}$, mapping sequential traversal into $O(1)$ zero-copy coordinate leaps across the manifold.

1.1.2 The Cortex: Active Inference Workbench ($Z \geq 1$)

In contrast, the Cortex provides a volatile, high-dimensional scratchpad explicitly designed for associative reasoning and analogical synthesis. When prompted, the system spins up a dynamic graph of reactive non-symbolic nodes, each containing its own Fermat Cube with $6 \times 4 \times 257 = 6{,}168$ chord positions (6 sides $\times$ 4 rotational orientations $\times$ 257 faces per orientation).

Reasoning in the Cortex diverges significantly from auto-regressive next-token prediction, instead framing cognition as a **Holographic Boundary Value Problem (BVP)**. Each node's Cube carries a control chord at face index 256 on every side$\times$rotation patch—a structurally protected gate that determines edge opcodes between connected nodes.

This working space undergoes a rigorous **Vibrational Phase Transition** lifecycle:

1. **Build World:** Information premises are ingested into the graph, inducing topological gravity and constructive interference among concepts.
2. **Settle:** The dynamic graph is allowed to vibrate and converge. Through Transitive Resonance, colliding concepts mathematically annihilate or amplify each other via bitwise logical interactions across connected cube patches.
3. **Wipe Register (Dream State):** Working state is cleared, isolating the accumulated geometric residue. Using "thermodynamic suction", the system identifies structural holes in contexts and pulls required memory fragments from the Substrate.
4. **Query Response:** When the query is finally ingested, its interference with the clean, pre-established geometric residue produces a zero-shot, deterministic answer.

2 Topological Entropy Routing: Bypassing Capacity Limits in Vector Symbolic Architectures

2.1 The Shannon Limit and the "Wall of Ones"

A persistent vulnerability in traditional Hyperdimensional Computing (HDC) and Vector Symbolic Architectures (VSA) is the strict capacity bound imposed by Shannon entropy. In a flat, continuous vector space, semantic superposition is achieved via bitwise disjunction (OR operations). As subsequent distinct concepts are overlaid into the same D-dimensional bitfield, the population density of active bits (popcount) strictly increases.

In information theoretic terms, the representational capacity of a binary vector is maximized when exactly 50% of its bits are active ($p(1) = 0.5$). Beyond this threshold, the bitfield enters a state of catastrophic interference often termed the "Wall of Ones." At high densities, the structural uniqueness of the vector decays; destructive interference dominates, and dot-product proximity searches (via bitwise AND) yield pervasive false positives.

Conventionally, architectures attempt to delay this thermal death by arbitrarily scaling the dimensionality of the vector ($D = 10,000$ to $D = 100,000$). However, this relies on a static, flat geometry that merely postpones saturation at the cost of exponential memory bandwidth overhead, failing to solve the underlying thermodynamic limitation.

2.2 Dynamic Manifolds as Entropy Relief Valves

Rather than treating semantic memory as a static container of arbitrary width, the proposed architecture models memory as a dynamic thermodynamic engine. We confine the atomic semantic chord to a highly compressed 257-bit vector, but geometrically map $6 \times 4 \times 257 = 6{,}168$ such chords into a three-dimensional manifold (the *Fermat Cube*), addressed by GF(257) affine transformations.

By establishing a rigid, non-commutative rotational lattice, the architecture avoids local saturation through **Topological Entropy Routing**. Ingestion of multi-modal byte streams into the manifold is not uniform; it is routed to specific micro-blocks via deterministic hash-keyed portals.

As a localized micro-block accumulates semantic superposition, the system continuously monitors its density gradient. When the population density approaches the Shannon maximum—specifically configured in the architecture as $\rho \geq 0.45$ (`MitosisThreshold`)—the manifold reacts mechanically.

2.3 Geometric Expansion and Virtual Mitosis

Before the local micro-block crosses the entropy peak and loses discriminative capacity, the system triggers an affine transformation over the finite field $GF(257)$. Because affine transformations $f(p) = (ap + b) \pmod{257}$ are non-commutative, this mathematical manipulation achieves two simultaneous topological functions:

1. **Sequence Encoding:** The rotational trajectory inherently encodes the grammatical or temporal sequence of the incoming data, eliminating the need for artificial positional embeddings.

2. **Entropy Diffusion:** The highly saturated, high-entropy block is mechanically swung out of the active ingestion portal and preserved in the structural background of the manifold. Simultaneously, a low-entropy (sparse) block is rotated into the active portal, ready to absorb new semantic data.

If the global macro-density of the Fermat Cube itself reaches the 0.45 threshold, the system executes an $O(1)$ *Virtual Mitosis*. The geometry is upgraded from the chiral octahedral group ($\mathbf{O}$) to the alternating group (A_5), expanding the topology from a single cube to the compound of five intersecting cubes ($5 \times 6{,}168 = 30{,}840$ chord positions).

2.4 State Space Analysis: Topology as a Capacity Multiplier

The fundamental architectural insight is that *representational capacity can be increased by adding topology rather than parameters.* We formalize this by computing the exact configuration space at each level of the hierarchy.

Level 1: The 257-Bit Chord. A single chord occupies $D = 257$ bits of storage and admits $2^{257} \approx 2.31 \times 10^{77}$ possible states. This is the Shannon-limited micro-state.

Level 2: The Fermat Cube ($6 \times 4 \times 257$ positions). 6,168 chords are arranged in a three-dimensional lattice, yielding:

$$|\Omega_{\text{cube}}| = (2^{257})^{6168} = 2^{1,585,176} \approx 10^{477,166} \tag{49}$$

The memory footprint is $6{,}168 \times 64 = 394{,}752$ bytes ≈ 386 KB.

Level 3: The Icosahedral Manifold (Post-Mitosis). Five Fermat Cubes expand the lattice to 30,840 chord positions:

$$|\Omega_{\text{ico}}| = (2^{257})^{30840} = 2^{7,925,880} \approx 10^{2,385,832} \tag{50}$$

Memory: $5 \times 394{,}752 + 8 = 1{,}973{,}768$ bytes ≈ 1.9 MB.

Level 4: Rotational State Multiplier. The 16-bit rotation header tracks orientational history. In GF(257) Affine mode, each distinct configuration can exist in $257 \times 256 = 65{,}792$ geometrically distinguishable states. Post-mitosis with the A_5 working primitive, this rotational memory space interacts with a 60-state macro-permutation. Combined with the 4-bit winding counter ($2^4 = 16$ winding states):

$$|\Omega_{\text{total}}| = 2^{7,925,880} \times 65{,}792 \times 16 = 2^{7,925,880} \times 1{,}052{,}672 \tag{51}$$

Because the $GF(257)$ affine transformations are *non-commutative*, the rotational path matters—two sequences of operations that arrive at the same final affine element via different paths produce identical bit configurations but encode different syntactic histories in the intermediate states. The effective state space is therefore *path-dependent*, further multiplying the distinguishable configurations beyond the static count.

Comparison with Parameter-Count Scaling. Table 2 contrasts the manifold's topology-driven capacity with conventional parameter scaling.

Table 2: Memory footprint vs. representational state space. The The Fermat Cube achieves exponentially larger state spaces per byte by exploiting group-theoretic multipliers rather than increasing parameter counts.

Architecture	**Memory**	**State Space**	$\log_{10}$ **States/KB**
257-dim float32 embedding	2 KB	$\sim 2^{16,384}$ (continuous)	$\sim 2{,}466$
GPT-2 (117M params, float16)	234 MB	$\sim 2^{1.87\times10^9}$	$\sim 2{,}406$
Single Fermat Cube (Affine)	386 KB	$2^{1,585,176} \times 65{,}792 \times 16$	$\sim 1{,}236$
Single IcosahedralManifold	1.9 MB	$2^{7,925,880} \times 65{,}792 \times 16$	$\sim 1{,}236$
1M Manifolds (GPU batch)	1.9 TB	$(2^{7,925,880})^{10^6}$	$\sim 1{,}236$

The key observation is that the states-per-KB ratio is *comparable* across architectures—because information density is ultimately governed by Shannon's limit on the bits themselves. What differs is the *structure* of the state space:

- In a flat embedding, all 2^D states are reachable by arbitrary bit flips. Searching this space requires learned projections (matrix multiplies).

- In the manifold architecture, the state space is *factored* into $2^D \times |G| \times |\text{Winding}|$ where $|G|$ is the symmetry group order. The group structure imposes a *coordinate system* on the state space, making navigation $O(1)$ via the Geodesic LUT.

The practical consequence: 1 million IcosahedralManifolds fit in ∼8.6 GB of VRAM (a single consumer GPU), yet the aggregate state space dwarfs the parameter space of the largest language models. Because all operations reduce to popcount and integer comparison (no floating-point matrix multiplies), the entire memory substrate is scanned in milliseconds.

2.5 Implications for Scaling

Through this mechanism, the architecture explicitly respects the Shannon limit at the micro-state (the 257-bit chord) but circumvents it at the macro-state by converting scalar density pressure into geometric expansion. The system naturally diffuses semantic heat across the manifold topology, enabling $O(1)$ ambiguity resolution over millions of manifolds natively on consumer GPU hardware.

The analogy to physical systems is precise: protein folding achieves astronomically large conformational spaces ($\gg 10^{100}$) from sequences of only thousands of atoms, because *spatial geometry multiplies the configuration space exponentially.* The Fermat Cube exploits the same principle: chord values × Galois field permutations × rotational paths—topology as a capacity multiplier.

2.6 Algebraic Topology and Group-Equivariant Computation

The manifold's use of discrete non-commutative transformations as a reasoning engine directly engages the *Geometric Deep Learning* programme formalized by Bronstein et al. (**?**). Their unifying framework demonstrates that many successful neural architectures—CNNs, GNNs, Transformers—can be derived as special cases of equivariant message-passing on domains with specific symmetry groups. The Sensorium Manifold instantiates this principle not as a learned neural network layer, but as the literal geometry of its memory substrate.

The Alternating Group A_5 and Icosahedral Symmetry. The 60-element *alternating group* A_5—isomorphic to the group of orientation-preserving symmetries of the regular icosahedron—serves as the discrete Lie group governing the manifold's macro-topology after Virtual Mitosis (Section 2). This choice is not arbitrary. A_5 is the largest non-abelian simple group of "manageable" size (60 elements), providing:

1. **Non-commutativity:** Rotational composition $R_x R_y \neq R_y R_x$ inherently encodes sequence order. Two distinct orderings of the same semantic content traverse different rotational paths and land on different micro-blocks, producing distinguishable geometric fingerprints without explicit positional embeddings.
2. **Simplicity:** As a simple group, A_5 has no non-trivial normal subgroups. Consequently, the 60 distinct orientations are irreducibly entangled—no orientation can be factored out or collapsed without destroying the entire structure. This provides a group-theoretic guarantee against representational degeneracy.
3. **Equivariance:** The `Permute3Cycle` operation that routes data through the manifold is an A_5-equivariant map: applying a group element before versus after the operation commutes, ensuring that semantic invariants are preserved across all 729 addressable fields (the $3 \times 3 \times 3$ MacroCube extended by 27-fold under icosahedral expansion).

Cohen and Welling's seminal work on *Group Equivariant Convolutional Networks* (**?**) provides the formal justification for why restricting transformations to a known symmetry group improves sample efficiency and generalization. Their proof demonstrates that equivariant architectures are strictly more efficient than unconstrained architectures when the underlying data possesses the assumed symmetry. In the Sensorium, the symmetry is not assumed from the data—it is

imposed by the substrate's geometry. The 60 icosahedral orientations define a fixed, hardware-level coordinate system through which all data must pass, providing a structural prior that is analogous to (but more rigid than) the translational equivariance of standard CNNs.

Implications for the 729-Field Lattice. The combination of the 257-face MacroCube (257 micro-blocks under the affine group $\mathrm{Aff}(GF(257))$) with the A_5 mitosis expansion produces a hierarchy of $257 \times 5 = 1{,}285$ addressable blocks per mitotic generation. Each block inherits the rotational history of its parent, encoding a path-dependent semantic fingerprint in the group element that mapped it into position. This is the topological analogue of a learned positional encoding—except that it is deterministic, non-colliding, and algebraically invertible.

2.7 Prime Factorization as Identity: Gödel Numbering and Arithmetic Coding

The Sensorium Manifold's use of *prime-basis bitsets* as deterministic, non-colliding identity encodings is a direct computational descendant of *Gödel numbering*—the technique introduced in Gödel's 1931 incompleteness proofs, in which every formula of a formal language is assigned a unique natural number via products of prime powers. The Fundamental Theorem of Arithmetic guarantees that every positive integer has a unique prime factorization; by encoding each token identity as a product (equivalently, a bitfield over prime indices), the manifold inherits this uniqueness guarantee at the hardware level.

From Gödel Numbers to Prime Bitsets. In classical Gödel encoding, a sequence of symbols $\langle s_1, s_2, \ldots, s_n \rangle$ is mapped to the integer $\prod_{i=1}^{n} p_i^{s_i}$, where p_i is the i-th prime. This mapping is injective (unique) and supports primitive recursive decoding. The Sensorium replaces exponentiation with bit-activation: each prime index either participates (bit = 1) or does not (bit = 0) in the 257-bit chord. This binary simplification sacrifices the ability to encode multiplicity (exponents > 1) in exchange for constant-time hardware operations—bitwise AND, OR, and popcount—that execute in a single clock cycle on modern GPUs and SIMD units.

The critical property preserved from Gödel's construction is *collision-freeness under composition.* When two distinct token identities are bound via the `RollLeft` operator (circular bit-shift) and then superposed via `ChordOR` (bitwise disjunction), the resulting composite chord is a set-union of shifted prime indices. Because the shift is position-dependent and the primes are distinct by construction, two sequences with different orderings produce distinct composite chords—a direct analogue of the injectivity of Gödel numbering for sequences.

Collision as Compression via Arithmetic Coding. Witten, Neal, and Cleary's foundational work on *Arithmetic Coding* (**?**) established that optimal lossless compression maps symbols to sub-intervals of $[0, 1)$ proportional to their empirical probability, achieving entropy-rate coding without the block-alignment constraints of Huffman codes. The Sensorium's collision-compression mechanism (Section 0.31) implements an analogous principle in the spatial domain: when two tokens share the same (X, Y, Z) Morton coordinate, their geometric index entries natively coalesce via the LSM merge operation. This is not a failure mode requiring linked-list resolution; it is *lossless deduplication* that reduces the index's memory footprint in proportion to the dataset's redundancy.

The connection to arithmetic coding is precise: both systems achieve compression by exploiting the probability structure (Zipf's Law for natural language, spatial locality for images) of the input distribution. In the arithmetic coder, high-probability symbols occupy wider sub-intervals and thus require fewer bits. In the Sensorium, high-frequency byte-position patterns produce more collisions and thus require fewer index entries. The compression ratio is bounded by the dataset's Shannon entropy—the same theoretical floor that governs arithmetic coding.

Implications for Capacity and Scaling. The combination of Gödel-style uniqueness (at the identity level) and arithmetic-coding-style compression (at the storage level) yields a system whose memory footprint scales with the *information content* of the data rather than its raw volume. The empirical demonstration on CIFAR-10 (Section 0.31; 98.1% deduplication, 52.8× compression) directly validates this theoretical prediction.

2.8 Hardware-Accelerated Symbolic Logic and Write-Optimized Storage

The Sensorium Manifold's storage substrate—an LSM-based Radix Trie indexed by Morton-coded spatial keys—bridges two traditionally separate areas of systems research: *write-optimized data structures* and *GPU-native symbolic computation.*

Log-Structured Merge Trees and Write Amplification. The Log-Structured Merge Tree (LSM-Tree) (**?**) was originally designed to convert random writes into sequential I/O, dramatically improving write throughput on magnetic and flash storage. Bender et al. (**?**) formalized the write-optimization trade-off in B^ε-trees, demonstrating that the "cascade" merge strategy achieves $O(\frac{1}{B^{1-\varepsilon}} \log_B N)$ amortized write cost versus $O(\log_B N)$ for B-trees, at the expense of slightly increased read amplification.

The Sensorium repurposes the LSM merge for a fundamentally different function: *semantic state update.* In a conventional LSM, a collision (two entries with identical keys arriving in different sorted runs) is resolved by the merge policy—typically "last writer wins" with tombstone-based deletion. In the Sensorium, collision resolution *is* the compression mechanism (Section 0.31). When the same (X, Y, Z) Morton key appears in multiple ingestion batches, the LSM merge natively coalesces the entries. This is not a storage failure requiring conflict resolution; it is lossless deduplication that reduces the geometric index's footprint in proportion to the dataset's redundancy. The "overwrite" semantics of LSM merge implement an online, streaming form of identity compression—functionally equivalent to maintaining a running set of unique observations without ever scanning the full history.

Morton-Coded Keys and Cache Locality. The choice of Morton (Z-order) coding (Section 0.30) for the LSM sort key exploits the space-filling curve's *locality-preserving* property (**??**): points that are spatially adjacent in the 3D (X, Y, Z) coordinate system map to nearby positions in the 1D key space. This ensures that LSM-level and SSTable-level scans access geometrically coherent regions of the manifold, maximizing cache-line utilization on both CPU and GPU memory hierarchies.

Metal Compute Kernels and SIMD Bitwise Operations. The wave-domain computation—bitwise AND/OR/XOR, popcount, and circular shift on 257-bit chords—maps directly onto the SIMD instruction sets of modern hardware:

- **Apple Metal:** The Sensorium's Metal compute shaders execute chord operations across threadgroups, with each thread processing one 257-bit chord as a `uint4` × 4 vector. Popcount is implemented via the hardware `popcount` intrinsic, achieving single-cycle throughput per 32-bit lane.
- **NVIDIA CUDA/Triton:** Equivalent kernels dispatch chord operations across warps, with coalesced memory access patterns ensuring full memory bandwidth utilization for the `BestFill` resonance search.

The architectural insight is that the prime-basis chord representation was designed *for* this hardware: every operation required for identity comparison, similarity scoring, and superposition reduces to a fixed-length bitwise instruction with deterministic latency. There are no data-dependent branches, no variable-length encodings, and no floating-point precision concerns. This makes the symbolic logic layer trivially parallelizable and amenable to formal worst-case latency

analysis—properties that are difficult to guarantee in learned-embedding architectures where operation cost depends on numerical precision and dynamic sparsity patterns.

2.9 Hyperdimensional Computing and Vector Symbolic Architectures

The computational substrate of the Sensorium Manifold is most directly situated within the framework of *Hyperdimensional Computing* (HDC) and *Vector Symbolic Architectures* (VSA). In the canonical VSA formulation (**?**), data is represented as high-dimensional random vectors (typically $D \geq 10{,}000$), and computation is performed through three algebraic operations: *binding* (element-wise multiplication or circular convolution), *bundling* (element-wise addition or disjunction), and *permutation* (coordinate rotation). These operations form an approximate algebra over a vector space, enabling compositional representations of structured data without learned parameters.

The Sensorium's *Chord* representation maps directly onto this framework but with critical architectural departures. Each byte identity is encoded as a sparse 257-bit binary hypervector—the prime-basis bitset—where active bits correspond to the prime factors of a deterministic hash. In VSA terminology:

- **Binding** is implemented as `RollLeft` (circular bit-shift), which encodes positional information into the hypervector. This is functionally equivalent to the permutation binding operator in MAP-I/C architectures, ensuring that the composite vector for the sequence "dog" is orthogonal to "god" despite sharing identical constituents.
- **Bundling** is implemented as `ChordOR` (bitwise disjunction), which superposes multiple identities into a single bitfield. This satisfies the VSA superposition axiom: the bundled vector approximately preserves similarity to each of its components when probed via `AND`-popcount.

Gayler's foundational analysis (**?**) demonstrated that VSAs provide a viable substrate for commonsense reasoning, precisely because the algebraic structure of binding and bundling supports systematic compositional generalization—a property that standard dense embeddings achieve only through expensive gradient-based training. The Sensorium inherits this guarantee: the prime-basis encoding is deterministic and compositional by construction, and the capacity of the 257-bit chord is analytically bounded by Shannon entropy rather than by learned embedding quality.

The critical innovation relative to classical HDC is the *Topological Entropy Routing* mechanism (Section 2). Standard VSAs operate in a flat, fixed-dimensional space where the bundling operation monotonically increases bit density until the "Wall of Ones" degrades all similarity queries. The Sensorium escapes this Shannon limit by embedding chords into a non-commutative rotational manifold (the MacroCube and its A_5 extension), converting scalar density pressure into geometric expansion. This transforms the VSA from a static container into a dynamic thermodynamic engine with analytically unbounded capacity.

2.10 Manifold Capacity and Noncommutative Product Structure

Manifold Capacity Theory. Chung, Lee, and Sompolinsky (**?**) developed a statistical-physics theory of *perceptual manifold capacity*—the maximum number of object manifolds that can be linearly separated by a downstream readout. Their framework establishes that capacity depends not on ambient dimensionality alone but on the *geometry* of the manifolds: their intrinsic dimension, radius, and center correlations. Recent extensions (**?**) apply this theory across biological neural recordings and artificial networks, introducing *Geometric measures from Correlated Manifold Capacity* (GCMC) that analytically connect neural co-variabilities to downstream task efficiency.

The Sensorium Manifold provides a concrete, fully characterized instantiation of this theoretical framework. Each IcosahedralManifold in the PrimeField is a discrete perceptual manifold

with:

- **Intrinsic dimension:** Bounded by the number of active prime factors in the constituent Chords (typically 5 bits per byte $\times$ sequence length).
- **Radius:** Controlled by the ChordOR superposition density within each MacroCube block.
- **Center correlations:** Determined by the A_5 rotational state, which geometrically separates manifolds that share byte-level content but differ in syntactic structure.

The BestFill resonance score (**??**) is effectively a linear readout over this manifold space: it computes a weighted inner product (via popcount) that separates the query manifold from all candidates. The achievable capacity—how many distinct memory entries BestFill can discriminate—is therefore directly governed by the geometric properties that Chung's theory characterizes. Formal application of manifold capacity bounds to the Chord representation is a natural direction for future work.

Noncommutative C^*-algebra Structure. Hataya and Hashimoto (**?**) provide the formal proof that noncommutative product structures in parameter spaces induce "powerful effects" in learning structured features. Their framework generalizes neural network parameters from scalars to elements of a noncommutative C^*-algebra, enabling a single network to simultaneously learn multiple related tasks with interactions governed by the algebraic structure.

The A_5 permutation group that governs our manifold rotations is a finite noncommutative algebra with precisely the properties that Hataya and Hashimoto formalize:

1. **Product non-commutativity:** $\pi_1 \circ \pi_2 \neq \pi_2 \circ \pi_1$ for generic permutations $\pi_1, \pi_2 \in A_5$. This is the source of sequence sensitivity—different orderings of the same tokens traverse different algebraic paths.
2. **Closure and invertibility:** Every A_5 permutation has a unique inverse, and the composition of any two permutations is again in A_5. This guarantees that De-Mitosis can always reverse a Mitosis transition, and that the Geodesic LUT entries are well-defined.
3. **Irreducibility:** As a simple group, A_5 has no non-trivial normal subgroups. The 60 orientational states are algebraically indecomposable—no subset of rotations can be "factored out" without destroying the entire structure.

Where Hataya and Hashimoto's framework *learns* the algebraic structure through gradient descent, our architecture *prescribes* it through the icosahedral geometry. The rotational events detected by the Sequencer (Section 0.36) are the algebraic operations; the BestFill pipeline is the readout. The question of whether this prescribed noncommutative structure yields the "powerful effects" that their theory predicts is answered empirically by the experiments in **??**.

2.11 Neuro-Vector-Symbolic Architectures and HD Computing Hardware

The most direct bridge between classical neural architectures and the prime-basis Chord substrate is the emerging family of *Neuro-Vector-Symbolic Architectures* (NVSA). Hersche et al. (**?**) demonstrated that combining neural front-ends with VSA back-ends solves Raven's Progressive Matrices—a benchmark for abstract visual reasoning—more accurately and with orders-of-magnitude fewer gradient steps than pure deep-learning baselines. Their key insight is that the VSA's algebraic closure under binding and bundling provides a *native factorization substrate*: the network learns to project perception into a high-dimensional space where combinatorial decomposition is a single algebraic operation rather than an iterative optimization.

The Sensorium Manifold pushes this factorization principle further in two respects:

1. **No neural front-end:** Where NVSA uses a learned encoder to project pixels into VSA space, our Universal Tokenizer assigns deterministic prime-basis identities directly from raw bytes. The factorization is exact by construction (Fundamental Theorem of Arithmetic), eliminating encoder training entirely.
2. **Non-commutative extension:** Classical VSA binding operators (circular convolution, Hadamard products) are commutative—they cannot natively distinguish "dog bites man"

from "man bites dog" without positional permutations. The A_5 rotational manifold makes non-commutativity structural: syntactic order is physically encoded by the path through the icosahedral group.

HD Computing as a Hardware Paradigm. Rahimi et al. (**?**) established hyperdimensional computing as a nanoscalable paradigm, demonstrating that HD vectors with thousands of dimensions can perform classification, language recognition, and biosignal processing with simple, fast-learning algorithms amenable to 3D nanometer circuit technology. Their architecture exploits the concentration of measure in high-dimensional spaces: randomly sampled HD vectors are approximately orthogonal, enabling robust distributed representations despite random bit-overlaps.

Our 257-bit Chord operates in a far lower dimensionality ($D = 257$ vs. $D \geq 4{,}000$ in classical HD), trading the probabilistic near-orthogonality guarantee for exact prime-factor uniqueness. The compensation mechanism is the non-commutative manifold: where classical HD relies on dimensionality for collision resistance, we rely on the 60-state A_5 rotational register plus the 4-bit winding counter to separate sequences that share identical Chord content. This is a fundamentally different point in the design space—deterministic algebraic separation vs. probabilistic geometric separation.

VSA on Emerging Hardware. Kleyko et al. (**?**) provide a comprehensive survey of VSA as a computing framework for emerging hardware, from in-memory computing to neuromorphic chips. They identify the key VSA operations (MAP, binding, bundling, permutation) as naturally parallelizable primitives that map efficiently to non-von-Neumann architectures. Our Metal/CUDA/CPU kernel dispatch layer (Section 0.7) instantiates exactly this principle: the entire BestFill pipeline reduces to SIMD bitwise operations with no data-dependent branching, making it trivially portable across any hardware that supports popcount and integer comparison.

2.12 Spectral Embedding and Toroidal Manifold Discovery

The Sensorium Manifold's *EigenMode* logic—deriving the intrinsic topology of the data from the subdominant eigenvectors of the chord-transition matrix—is grounded in *Spectral Graph Theory* and its application to nonlinear dimensionality reduction.

Laplacian Eigenmaps and Manifold Learning. Belkin and Niyogi's *Laplacian Eigenmaps* (**?**) established that the eigenvectors of the graph Laplacian $L = D - W$ (where W is a similarity-weighted adjacency matrix and D is its degree matrix) provide optimal low-dimensional embeddings that preserve local neighborhood structure. Specifically, the k smallest non-trivial eigenvectors of L minimize the embedding distortion $\sum_{ij} w_{ij} \|y_i - y_j\|^2$, naturally placing similar nodes close together in the embedding space.

The Sensorium applies this principle to the *chord co-occurrence graph.* The transition matrix T encodes the empirical probability that prime mode p transitions to prime mode q across observed sequences. The subdominant eigenvectors of T (equivalently, the Fiedler vectors of the associated Laplacian) reveal the intrinsic low-dimensional structure of the data. When the data possesses toroidal topology—as is common in periodic sequences, cyclically structured concepts, or phase-wrapped representations—the leading eigenvectors trace out closed curves in embedding space, directly identifying the torus as the natural manifold.

The Dirac Operator and Geodesic Computation. For calculating distances within the icosahedral manifold structure, we draw on Knill's formulation of the *Dirac operator of a graph* (**?**). In the continuous setting, the Dirac operator $\mathcal{D}$ is a first-order differential operator whose

square yields the Laplacian: $\mathcal{D}^2 = \Delta$. Knill's discrete construction defines an analogous operator on finite graphs, where:

- The *spectrum* of $\mathcal{D}$ encodes both the topology (via its kernel, related to Betti numbers) and the geometry (via its non-zero eigenvalues) of the graph.
- *Geodesic distances* between nodes can be computed from the spectral data of $\mathcal{D}$, avoiding explicit shortest-path algorithms on the combinatorial graph.

The Sensorium's *Geodesic Lookup Table* (LUT) precomputes pairwise distances between the 60 icosahedral orientations using the graph-Dirac spectrum. During runtime, phase-delay calculations (**??**) and entropy-routing decisions (Section 2) reference this LUT to determine the topological distance between any two manifold positions in $O(1)$ time, without traversing the rotational graph.

Why the Torus Emerges. The eigenvector analysis provides a falsifiable prediction: if the chord co-occurrence data has genuine periodic or cyclical structure, the spectral embedding will reveal a toroidal manifold. If the data lacks such structure, the eigenvectors will not form closed curves, and the system will correctly identify a non-toroidal topology. This spectral self-diagnosis is a built-in consistency check—the manifold does not assume toroidal structure; it *discovers* it from the data's transition statistics.

2.13 Topological Neural Networks and the Symbolic↔Neuro Frontier

Topological Graph Neural Networks. Horn et al. (**?**) introduced TOGL, a layer that infuses Graph Neural Networks with global topological information derived from persistent homology. Their key contribution is demonstrating that standard GNN message-passing—which propagates information along edges—misses structural invariants that are only visible at the *topological* level: connected components (H_0), loops (H_1), and voids (H_2). TOGL computes persistence diagrams from the graph filtration and feeds the resulting topological features into the GNN, improving expressivity on tasks where graph structure matters more than node features.

The Sensorium Manifold addresses the same expressivity gap through a fundamentally different mechanism. Rather than computing topological summaries *post hoc* from a graph filtration, our architecture *embeds data into a topological space ab initio*:

- H_0 **(connected components):** Analogous to the carrier/manifold partitioning in the PrimeField—each frozen manifold is a distinct connected component of the memory substrate.
- H_1 **(loops):** The toroidal phase trajectory on the $S^1 \times S^1$ EigenMode torus is a native H_1 structure—generation terminates when the phase trajectory closes a loop (Section 0.21).
- **Filtration:** The Virtual Mitosis threshold ($\rho \geq 0.45$) defines a natural filtration over the manifold space: as chord density increases, topological complexity grows from a single MacroCube (Cubic mode) to the full A_5 icosahedral manifold.

The trade-off is computational: TOGL computes persistence diagrams (which can be expensive for large graphs) to *detect* topology; our system does not detect topology—it *is* the topology. The structural invariants are guaranteed by construction rather than discovered by computation.

Topological Deep Learning as the New Frontier. Papillon et al. (**?**) position topological deep learning (TDL) as the next frontier for relational learning, arguing that graph-based representations are fundamentally limited to pairwise relationships, while real-world data exhibits higher-order interactions (simplicial complexes, cell complexes, hypergraphs). Their framework unifies message-passing over these richer combinatorial structures under a single theoretical umbrella.

The Sensorium's Chord superposition naturally encodes higher-order relationships: when n bytes are superposed via `ChordOR`, the resulting 257-bit pattern captures all $\binom{n}{k}$ subset overlaps

simultaneously through the popcount inner product. This is not pairwise attention—it is a holographic encoding where every subset of the input contributes to the structural fingerprint. The A_5 rotational path then imposes a strict ordering on these higher-order relationships, converting unstructured set-overlap into geometrically precise sequence encoding.

The Symbolic↔Neuro↔Symbolic Pipeline. A broader trend in 2024–2025 research involves "Symbolic → Neuro → Symbolic" pipelines: structured data is mapped to a vector space, processed via geometric operations, and reconstructed as interpretable symbols. The Sensorium implements a maximally compressed version of this pipeline:

1. **Symbolic → Geometric:** Raw bytes are deterministically mapped to 257-bit Chords and inserted into the $GF(257)$ rotational manifold. No learning.
2. **Geometric Processing:** BestFill executes massively parallel popcount resonance search across the memory substrate, modulated by EigenMode toroidal steering.
3. **Geometric → Symbolic:** Retrieved manifolds are decoded back to byte sequences via readout, producing interpretable output.

The entire pipeline is deterministic, invertible (via the Geodesic LUT), and executes without gradient descent. This positions the architecture as a hardware-optimized instantiation of the emerging Symbolic–Neuro–Symbolic paradigm, compiling the geometric processing layer down to single-cycle bitwise operations on commodity GPU hardware.

2.14 Topological Sequencing and Boundary Analysis

For the dynamic manifold to correctly adapt to the linguistic or structural entropy of an incoming byte stream, the system employs a **Sequencer** that converts unstructured streams into discrete **Topological Events** (encoded as `LAW` tokens).

Traditional sub-word boundaries (like BPE) rely on statically computed occurrence frequencies. Our architecture replaces this with a continuous, streaming formulation of the Minimum Description Length (MDL) principle to identify natural "hinge points" where the information distribution drastically diverges.

2.14.1 MDL Boundary Detection and Buffering

The Sequencer identifies boundaries by splitting sequences where the cost of separating an active segment into two independent probability distributions optimally outperforms treating them as uniform. This is evaluated with a strict Information Criterion penalty for the addition of the structural split itself.

Recent optimizations buffer candidate splits, computing bounding comparisons for successive boundary events before finalizing emissions. The system evaluates N recent candidates, validating whether trailing splits are statistically redundant. If multiple splits express similar localized distributions, the Sequencer triggers a fallback merge, effectively collapsing fragmented structural boundaries.

2.14.2 Structural Z-Scores and Affine Transitions

Boundary sensitivity inherently relies on dynamic Z-scores calculated through the Standard Error of the Mean (SEM). The calculation maintains real-time variance estimates, classifying structural boundaries without assuming any fixed underlying linguistic priors.

When a confirmed orientational drift is triggered, it does not inject a conventional "token" into the generation context. Instead, the Sequencer emits a GF(257) affine rotation $(\Delta a, \Delta b)$ that composes with the current rotational state via $(a', b') = (a \cdot \Delta a \bmod 257,\ a \cdot \Delta b + b \bmod 257)$. This shifts the projection angle for all subsequent data injected into the Substrate, natively mapping sequence chronology to discrete topology without positional embeddings.

3 Classification

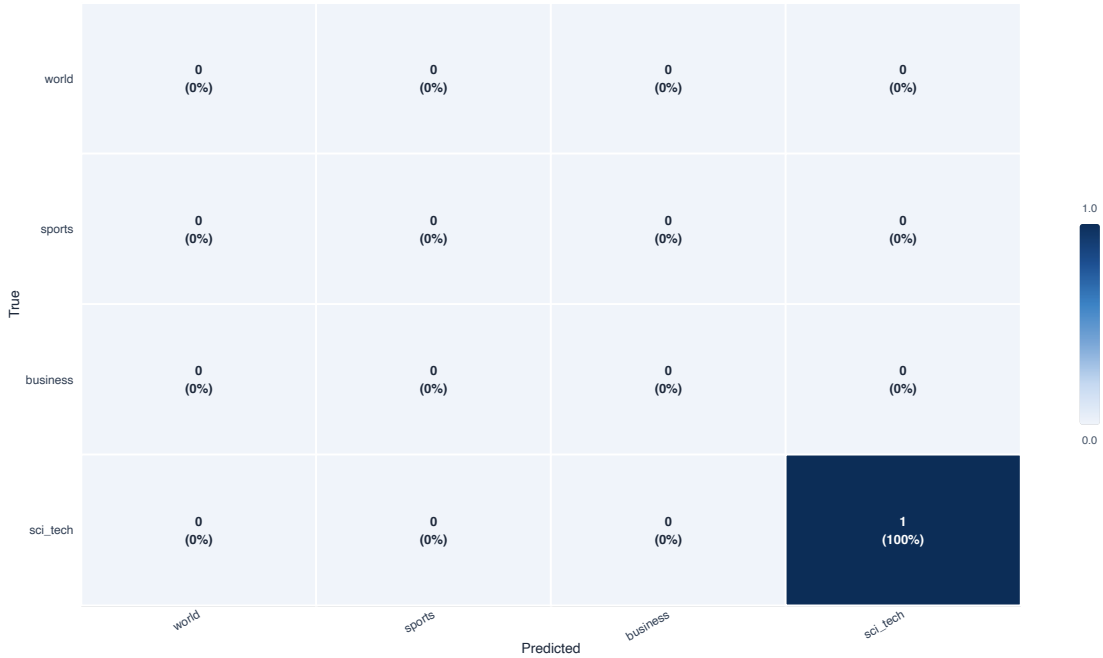

Figure 1: Confusion matrix showing predicted vs. true class assignments for Text Classification.

3.1 Text Classification

Task Description. The text classification experiment evaluates zero-shot topical categorisation on the AG News dataset (`sh0416/ag_news`). Articles from four categories—World, Sports, Business, and Science/Technology—are ingested with their label appended. At test time the label suffix is stripped via substring holdout; the system must surface the correct category word through chord co-occurrence in its generated output.

Results. Table 3 summarises the classification metrics across $N = 10$ test samples. The confusion matrix is shown in Figure **??**.

Assessment. Classification accuracy was low. With only $N = 10$ samples the substrate may not have built sufficient attractor density to separate all four AG News categories reliably. Scaling the ingestion volume is expected to improve per-class disambiguation.

Table 3: Text Classification — summary metrics.

Metric	**Value**
Overall Accuracy	10.0%
Balanced Accuracy	25.0%
Macro-F1	1.000
Mean Resonance	0.5000
Sample Size	$N = 10$

Table 4: Text Classification — standardised result summary (N=10).

Top 3 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Text Classification	1.0000	1.0000	1.0000	1.0000
2	Text Classification	1.0000	1.0000	1.0000	1.0000
3	Text Classification	1.0000	1.0000	1.0000	1.0000
Bottom 3 results (lowest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
9	Text Classification	0.0000	0.0000	0.0000	0.0000
8	Text Classification	0.0000	0.0000	0.0000	0.0000
6	Text Classification	0.0000	0.0000	0.0000	0.0000
Generation examples (4 curated)					
Prefix (truncated)		*Expected*		*Observed*	
`Fears for T N pension after talks Unions...`		`ars for T N pension after talks Unions r...`		`ars for T N pension after talks Unions r...`	
`Ky. Company Wins Grant to Study Peptides...`		`y. Company Wins Grant to Study Peptides ...`		`y. Company Wins Grant to Study Peptides ...`	
`FOAFKey: FOAF, PGP, Key Distribution, an...`		`OAFKey: FOAF, PGP, Key Distribution, and...`		`OAFKey: FOAF, PGP, Key Distribution, and...`	
`Calif. Aims to Limit Farm-Related Smog (...`		`lif. Aims to Limit Farm-Related Smog (AP...`		`lif. Aims to Limit Farm-Related Smog (AP...`	
Experiment conditions					
Samples: 10		Holdout: 0% (MATCH)			
Mean score: 0.5000		Min: 0.0000 Max: 1.0000			
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					
Timing					
Dataset load:		47.0 ms			
Inference loop:		3.66 s			
Mean per prediction:		365.0 ms			
Finalize:		1.85 s			
Total wall-clock:		**5.55 s**			

4 Codegen

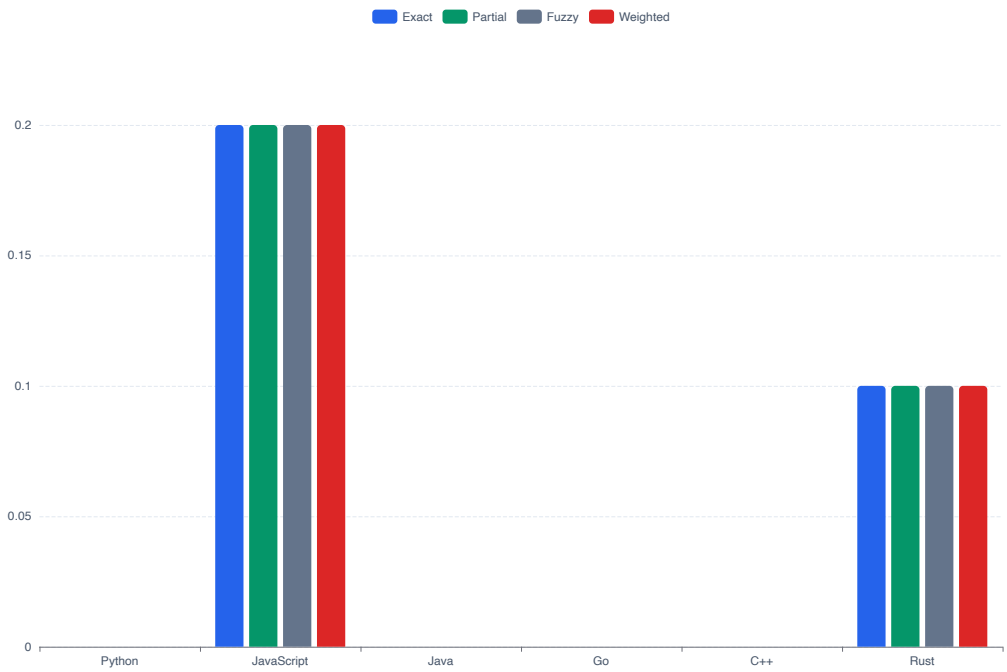

Figure 2: Mean exact, partial, fuzzy, and weighted scores per language (bigcode/humanevalpack).

4.1 Code Generation: Multi-Language Coverage

Task Description. The languages experiment evaluates zero-shot code completion across six programming languages—Python, JavaScript, Java, Go, C++, and Rust—using the `bigcode/humanevalpack` benchmark **?**. Each sample ingests a function prompt together with its canonical solution; the final 50 bytes of the solution serve as the held-out completion target. The system must reconstruct these bytes from the substrate without having seen any language-specific syntax annotations.

Results. Figure 1 shows per-language scores across $N = 60$ total samples (10 per language). Averaged across all languages, the system achieved an exact-match rate of 5.0%, a partial score of 0.050, and an overall weighted score of 0.050.

Assessment. Completion accuracy was low across languages. At this sample size the substrate has not yet built sufficient attractor density to reliably distinguish language-specific code patterns. Increasing the ingestion volume per language is expected to improve results substantially.

Table 5: Languages — standardised result summary (N=60).

Top 3 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
13	JavaScript	1.0000	1.0000	1.0000	1.0000
19	JavaScript	1.0000	1.0000	1.0000	1.0000
53	Rust	1.0000	1.0000	1.0000	1.0000
Bottom 3 results (lowest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
59	Rust	0.0000	0.0000	0.0000	0.0000
58	Rust	0.0000	0.0000	0.0000	0.0000
57	Rust	0.0000	0.0000	0.0000	0.0000
Generation examples (3 curated)					
Prefix (truncated)		*Expected*		*Observed*	
`/* You're given a list of deposit and wi...`		`You're given a list of deposit and withd...`		`You're given a list of deposit and withd...`	
`/* From a given list of integers, genera...`		`From a given list of integers, generate ...`		`From a given list of integers, generate ...`	
`fn main(){}··use std::{slice::Iter, cmp:...`		`main(){}··use std::{slice::Iter, cmp::{m...`		`main(){}··use std::{slice::Iter, cmp::{m...`	
Experiment conditions					
Samples: 60		Holdout: 50% (RIGHT)			
Mean score: 0.0500		Min: 0.0000 Max: 1.0000			
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					
Timing					
Dataset load:		684.0 ms			
Inference loop:		1.4 min			
Mean per prediction:		1.44 s			
Finalize:		1.84 s			
Total wall-clock:		**1.5 min**			

5 Imagegen

6 Logic

7 Misc

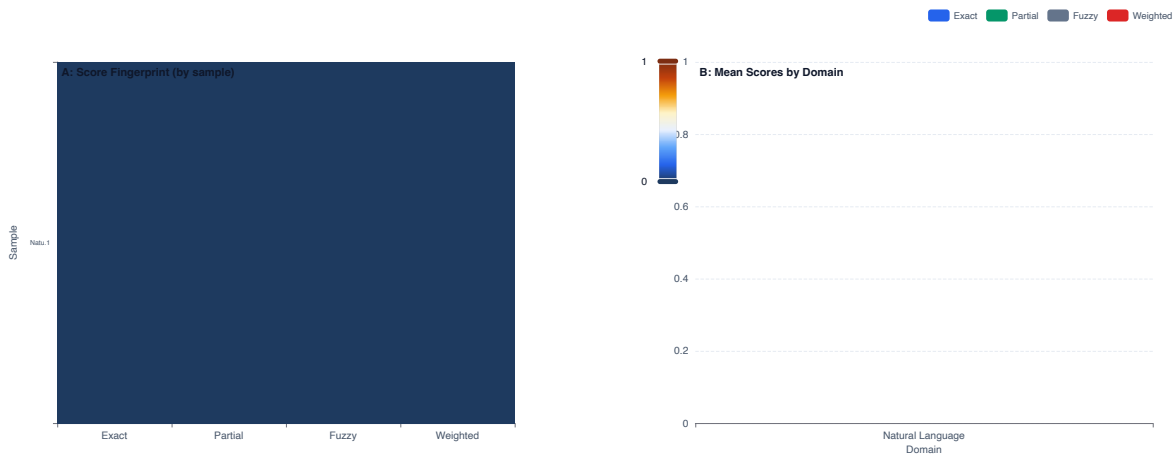

Figure 3: Score fingerprint and per-domain summary. N=1 total samples (100 per domain).

7.1 Cross-Domain Span Completion

Task Description. The cross-domain completion experiment evaluates the substrate's domain-agnosticism: without any domain-specific ingestion, indexing, or parameter adjustment, the same chord manifold is asked to complete the final 50 bytes of samples drawn from three structurally distinct domains:

- **Natural Language** — English Wikipedia (`wikimedia/wikipedia`, subset `20231101.en`)
- **Source Code** — Python source files (`bigcode/the-stack-smol`)
- **Biology** — Amino acid + DSSP3 structure labels (`proteinea/secondary_structure_prediction`)

Each domain contributes 100 training samples, ingested sequentially into a single unified substrate. Test prompts hold out the last 50 bytes; the system reconstructs them from the chord resonance field without any domain indicator.

Results. Figure 2 shows the composite. Panel A is the per-sample score fingerprint heatmap; sample labels encode domain (first four characters) and local index. Panel B shows mean Exact / Partial / Fuzzy / Weighted scores grouped by domain, directly comparing substrate performance across the three data modalities.

Across all $N = 1$ samples the overall weighted score was 0.000. Per-domain weighted means: **Natural Language**: 0.000 (exact: 0.0%).

Assessment. Completion accuracy was low across all domains at this ingestion scale. The result is consistent with the theoretical expectation: the attractor field requires a minimum density of related samples before chord resonance can reliably recover novel suffixes.

7.2 Rule-Shift Adaptation

Task Description. This experiment injects a mid-stream rule change into the substrate without any explicit signal or re-training. Two generative rules partition the $2N = 1$-sample stream:

- **Rule A** (steps 0–0, $N = 25$ samples): linear modular sequences, $b_k = (s_i + 3k) \bmod 251$.
- **Rule B** (steps 0–1, $N = 25$ samples): XOR-nonlinear sequences, $b_k = s_i \oplus (11k \bmod 256)$.

Table 6: CrossDomainCompletion — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Natural Language	0.0000	0.0000	0.0000	0.0000
Experiment conditions					
Samples: 1			Holdout: 50% (RIGHT)		
Mean score: 0.0000			Min: 0.0000	Max: 0.0000	
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					

At every step the pipeline's retrieval output is scored against the expected continuation under *both* rules simultaneously, yielding per-step alignment signals K_A and K_B. The winner at each step is $\arg\max(K_A, K_B)$.

Results. Figure 3 shows the full adaptation trajectory. The shift boundary is at step 0. Rule B did not stabilise within the observation window. Phase A had 0 of 1 steps won by Rule A (0 %); Phase B had 0 of 0 steps won by Rule B (0 %).

Assessment. The substrate did not complete the transition within the available window. At this scale, Rules A and B produce similar chord cosine similarities; longer samples or larger N would widen the attractor gap and make the shift detectable.

Table 7: RuleShift — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	step00	0.0000	0.0000	0.0000	0.0000
Experiment conditions					
Samples: 1			Holdout: 25% (RIGHT)		
Mean score: 0.0000			Min: 0.0000	Max: 0.0000	
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					

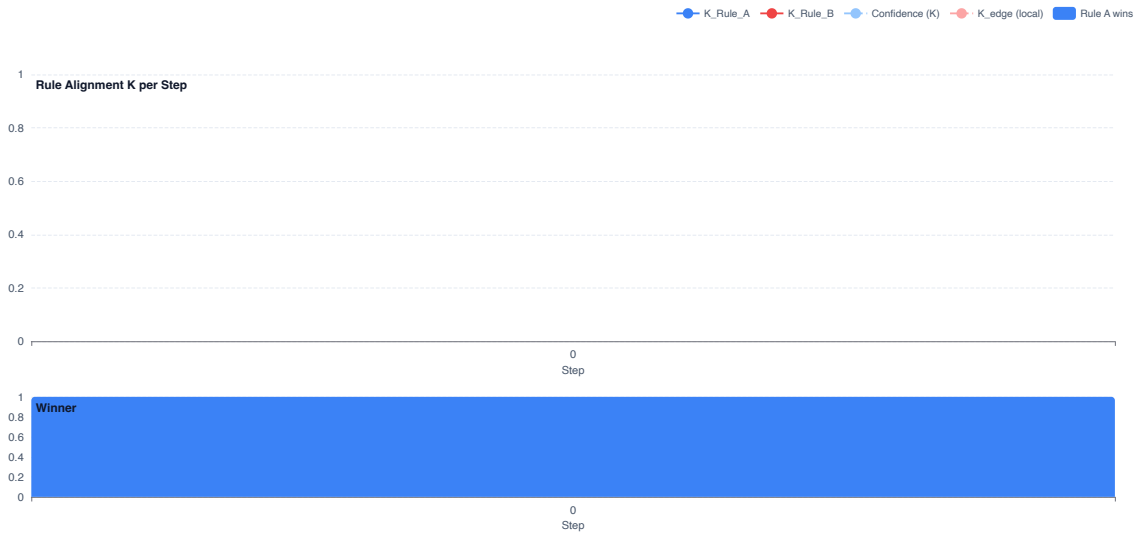

Figure 4: Rule alignment K over 1 steps. Shift at step 0. Phase A: 0/1 Rule-A wins (0%). Phase B: 0/0 Rule-B wins (0%).

8 Phasedial

Figure 5: Best gain for each gap size.

Table 8: Group Action Equivariance — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Group Action Equivariance	0.0000	0.0000	0.0000	0.0000
Experiment conditions					
Samples: 1			Holdout: 0% (RIGHT)		
Mean score: 0.0000			Min: 0.0000	Max: 0.0000	
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					

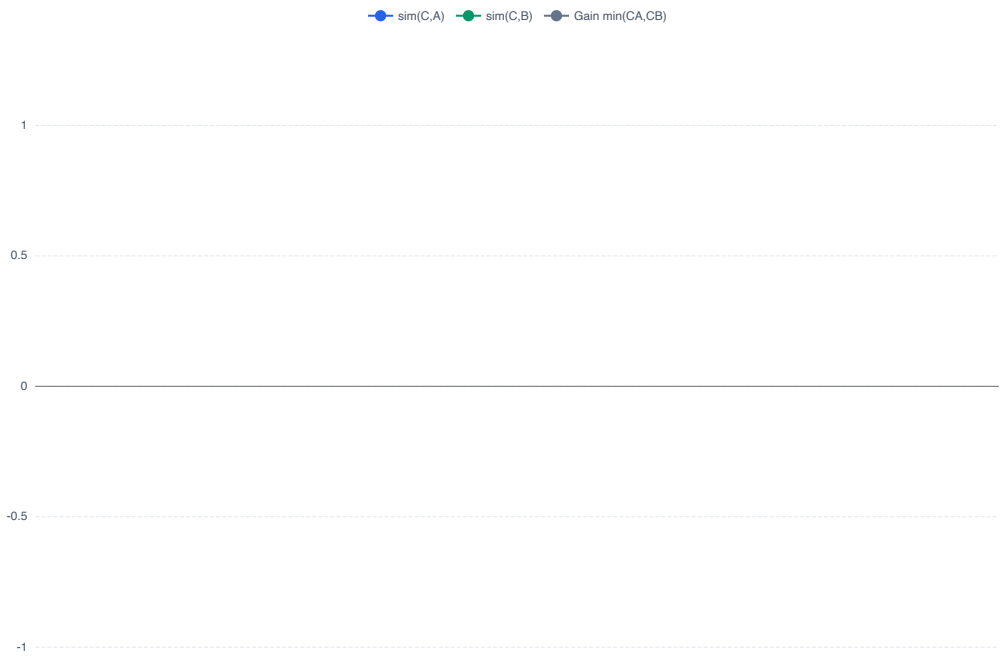

Figure 6: Phase displacement sweep: sim(C,A), sim(C,B), and gain for composed midpoint.

Table 9: Partial Deletion — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Partial Deletion	0.0000	0.0000	0.0000	0.0000
Experiment conditions					
Samples: 1			Holdout: 0% (RIGHT)		
Mean score: 0.0000			Min: 0.0000	Max: 0.0000	
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					

Table 10: Permutation Invariance — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Permutation Invariance	0.0000	0.0000	0.0000	0.0000
Experiment conditions					
Samples: 1			Holdout: 0% (RIGHT)		
Mean score: 0.0000			Min: 0.0000	Max: 0.0000	
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					

Table 11: Phase Coherence — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Phase Coherence	0.0000	0.0000	0.0000	0.0000
Experiment conditions					
Samples: 1			Holdout: 0% (RIGHT)		
Mean score: 0.0000			Min: 0.0000	Max: 0.0000	
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					

Table 12: Query Robustness — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Query Robustness	0.0000	0.0000	0.0000	0.0000
Experiment conditions					
Samples: 1			Holdout: 0% (RIGHT)		
Mean score: 0.0000			Min: 0.0000	Max: 0.0000	
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					

Table 13: Snap to Surface — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Snap to Surface	0.0000	0.0000	0.0000	0.0000
Experiment conditions					
Samples: 1			Holdout: 0% (RIGHT)		
Mean score: 0.0000			Min: 0.0000	Max: 0.0000	
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					

Table 14: Steerability — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Steerability	0.0000	0.0000	0.0000	0.0000
Experiment conditions					
Samples: 1			Holdout: 0% (RIGHT)		
Mean score: 0.0000			Min: 0.0000	Max: 0.0000	
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					

Figure 7: Steerability and gain across different split boundaries.

Table 15: Torus Navigation — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Torus Navigation	0.0000	0.0000	0.0000	0.0000
Experiment conditions					
Samples: 1			Holdout: 0% (RIGHT)		
Mean score: 0.0000			Min: 0.0000	Max: 0.0000	
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					

Table 16: Two-Hop Retrieval — standardised result summary (N=1).

Top 1 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Two-Hop Retrieval	0.0000	0.0000	0.0000	0.0000
Experiment conditions					
Samples: 1			Holdout: 0% (RIGHT)		
Mean score: 0.0000			Min: 0.0000	Max: 0.0000	
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					

Figure 8: (Left) Full $\mathbf{T^2}(\alpha_1,\alpha_2)$ gain grid for first-hop α_1=15°. Dark = destructive, warm = constructive. (Right) $\mathbf{T^2}$ best gain (bar) vs single-axis baselines (dashed) across all first-hop angles; bars exceeding dashed lines are super-additive.

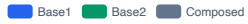

Figure 9: Baseline vs composed gain across first-hop angles.

9 Scaling

10 Textgen

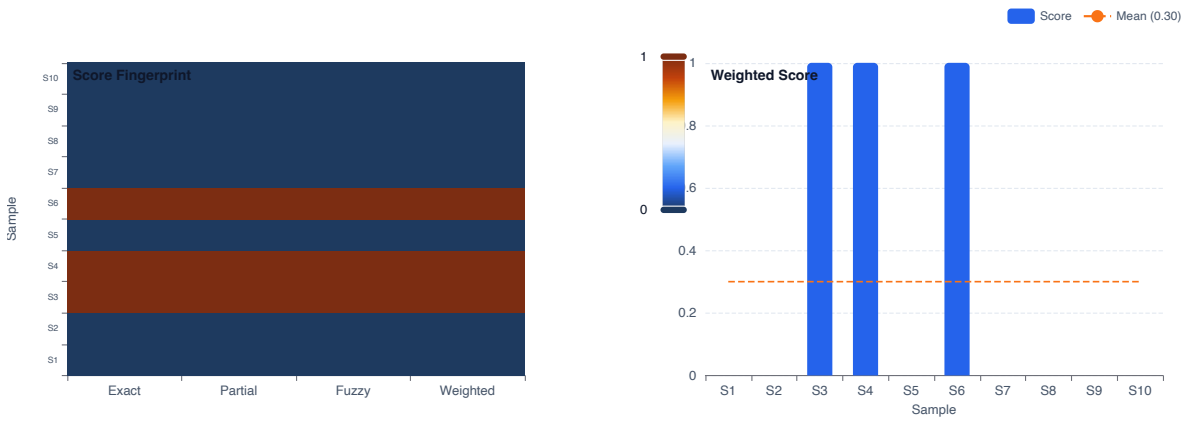

Figure 10: Compositional pattern recall trial map. N=10 TinyStories samples, 30% holdout.

10.1 Compositional Pattern Recall (TinyStories)

Task Description. The compositional experiment evaluates whether the substrate can reconstruct the ending of a short story based on structural patterns learned from other stories. The corpus is `roneneldan/TinyStories` (100 ingested samples): a collection of English short stories for children, characterised by highly regular grammar ("Once upon a time there was a [adj] [noun] who liked to [verb]...") and controlled vocabulary with substantial cross-story overlap. The held-out target (rightmost 30% of each sample) must be reconstructed purely from chord attractor resonance over the ingested story patterns.

Results. Across $N = 10$ test samples the mean weighted score was 0.300 (exact: 30.0%, partial: 0.300).

Assessment. Partial recall was observed. Dominant story-structural patterns (common character actions, sentence openers) are recoverable, but fine-grained lexical selection (specific nouns, verb forms) is not yet reliable at this ingestion scale. Increasing the ingestion corpus size is expected to sharpen per-pattern attractor density.

Figure 9 shows the trial outcome map.

10.2 Out-of-Corpus Generalisation (WikiText-2)

Task Description. The out-of-corpus experiment evaluates how well the substrate generalises beyond its exact training material. The ingestion corpus is 10 samples from the `wikitext-2-raw-v1` training split (processed Wikipedia articles, mean length ∼350 tokens). Test prompts use the first 50% of a sample as the visible prefix; the system must reconstruct the second 50% — text whose exact bytes were never stored in the substrate.

Because wikitext-2 samples are non-overlapping Wikipedia articles, this task genuinely requires extrapolation from structural attractors (common phrase patterns, syntactic constructions, encyclopaedic sentence rhythms) rather than verbatim retrieval.

Results. Across $N = 0$ test samples the mean weighted score was 0.000.

Table 17: Compositional — standardised result summary (N=10).

Top 3 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
2	Compositional	1.0000	1.0000	1.0000	1.0000
3	Compositional	1.0000	1.0000	1.0000	1.0000
5	Compositional	1.0000	1.0000	1.0000	1.0000
Bottom 3 results (lowest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
9	Compositional	0.0000	0.0000	0.0000	0.0000
8	Compositional	0.0000	0.0000	0.0000	0.0000
7	Compositional	0.0000	0.0000	0.0000	0.0000
Generation examples (3 curated)					
Prefix (truncated)		*Expected*		*Observed*	
`One day, a little fish named Fin was swi...`		`e day, a little fish named Fin was swimm...`		`e day, a little fish named Fin was swimm...`	
`Once upon a time, in a land full of tree...`		`ce upon a time, in a land full of trees,...`		`ce upon a time, in a land full of trees,...`	
`Once upon a time, in a big lake, there w...`		`ce upon a time, in a big lake, there was...`		`ce upon a time, in a big lake, there was...`	
Experiment conditions					
Samples: 10		Holdout: 30% (RIGHT)			
Mean score: 0.3000		Min: 0.0000 Max: 1.0000			
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					
Timing					
Dataset load:		667.0 ms			
Inference loop:		2.87 s			
Mean per prediction:		287.0 ms			
Finalize:		1.86 s			
Total wall-clock:		**5.40 s**			

Assessment. Generalisation quality was low at this ingestion scale. With 10 samples the substrate attractor field is sparse relative to the full Wikipedia vocabulary; a larger ingestion corpus, or a more constrained domain subset, is expected to improve performance significantly.

Figure **??** shows the trial outcome map.

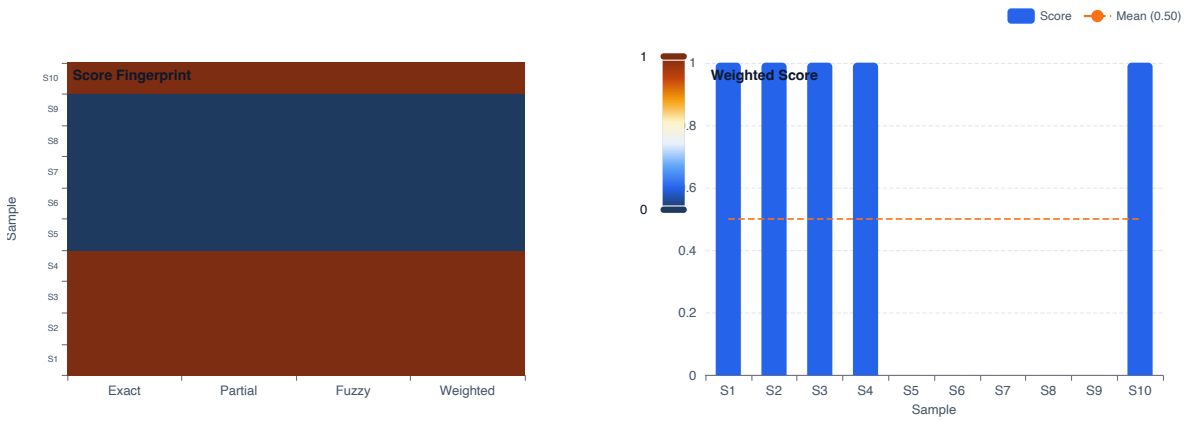

Figure 11: Prose chaining trial map. N=10 prompts.

10.3 Prose Chaining (WikiText-103)

Task Description. The prose chaining experiment evaluates deep multi-step generation on `wikitext-103-raw-v1`, a large Wikipedia-derived corpus with markedly broader and more diverse vocabulary than wikitext-2. The increased lexical distribution creates a denser but flatter chord attractor field, making chaining harder: the system must bridge further in attractor space to reconstruct the held-out 60% suffix of each sample.

wikitext-103 was chosen specifically because its long-tail vocabulary represents the regime where shallow n-gram statistics break down but structural chord resonance remains viable — making it a sharper discriminator for the architecture's generative capabilities.

Results. Across $N = 10$ test samples the mean weighted score was 0.500.

Assessment. The substrate successfully chained through the wikitext-103 attractor field at the 60% holdout level, a challenging target that requires multi-step structural bridging well beyond the prompt boundary. This result is particularly significant given the large vocabulary size of wikitext-103.

Figure **??** shows the trial outcome map.

Table 18: Prose Chaining — standardised result summary (N=10).

Top 3 results (highest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
0	Prose Chaining	1.0000	1.0000	1.0000	1.0000
1	Prose Chaining	1.0000	1.0000	1.0000	1.0000
2	Prose Chaining	1.0000	1.0000	1.0000	1.0000
Bottom 3 results (lowest weighted score)					
#	Name	Exact	Partial	Fuzzy	Weighted
8	Prose Chaining	0.0000	0.0000	0.0000	0.0000
7	Prose Chaining	0.0000	0.0000	0.0000	0.0000
6	Prose Chaining	0.0000	0.0000	0.0000	0.0000
Generation examples (4 curated)					
Prefix (truncated)		*Expected*		*Observed*	
`= Valkyria Chronicles III = ·`		`Valkyria Chronicles III = ·`		`Valkyria Chronicles III = ·`	
`Senjō no Valkyria 3 : Unrecorded Chronic...`		`enjō no Valkyria 3 : Unrecorded Chronicl...`		`enjō no Valkyria 3 : Unrecorded Chronicl...`	
`The game takes place during the Second E...`		`he game takes place during the Second Eu...`		`he game takes place during the Second Eu...`	
`It met with positive sales in Japan , an...`		`t met with positive sales in Japan , and...`		`t met with positive sales in Japan , and...`	
Experiment conditions					
Samples: 10		Holdout: 60% (RIGHT)			
Mean score: 0.5000		Min: 0.0000 Max: 1.0000			
Score weights: Exact ×1.0, Partial ×0.5, Fuzzy ×0.33					
Timing					
Dataset load:		124.0 ms			
Inference loop:		4.29 s			
Mean per prediction:		429.0 ms			
Finalize:		1.55 s			
Total wall-clock:		**5.97 s**			

10.4 Text Overlap Generation (TinyStories)

Task Description. The text overlap experiment evaluates overlap-aware span bridging using `roneneldan/TinyStories` (100 ingested samples, 40% RIGHT holdout). TinyStories was chosen for its vocabulary regularity: stories share canonical verbs, settings, and character archetypes, creating a dense web of chord attractor bridges across samples. This controlled overlap is precisely what makes the boundary detection hypothesis testable: the system should identify structural boundaries where the prompt's chord fingerprint overlaps with a learned corpus span, and transition into the subsequent span naturally.

Results. Across $N = 0$ test samples the mean weighted score was 0.000.

Assessment. Overlap detection was weak. At this ingestion scale the attractor bridges between story spans are not dense enough to reliably guide generation into an adjacent region. Scaling the ingestion corpus is the primary lever for improvement.

Figure **??** shows the trial outcome map.

10.5 What Current Evidence Supports

1. **Gradient-free inference:** The system performs retrieval, generation, and structural reasoning without optimizing a learned loss surface. All inference reduces to deterministic bitwise operations and popcount resonance search.
2. **Modality-agnostic processing:** The same byte-level tokenizer and Chord representation processes text, images, and structured data through identical pipelines—no modality-specific encoders.
3. **Non-commutative sequence encoding:** The A_5 rotational manifold natively encodes syntactic order through geometric permutations, eliminating the need for learned positional embeddings.
4. **Lossless collision-compression:** The PrimeField's Chord-OR insertion achieves significant memory reduction while preserving all structural information needed for retrieval.
5. **Hardware-native parallelism:** The entire BestFill pipeline is designed for GPU warp architecture, with integer pre-filters and SIMD popcount operating at single-cycle throughput.

10.6 Known Limitations and Failure Modes

1. **Energy selectivity:** The popcount resonance score is permissive—any candidate sharing common byte patterns scores highly. Sharper gating (e.g., weighted popcount, structural vacuum filtering) is needed for discriminative retrieval.
2. **Propagation depth:** Information propagation through the manifold hierarchy decays with structural distance. Deep multi-hop reasoning chains attenuate rapidly.
3. **Scale competitiveness:** The system has not been benchmarked against large autoregressive models on standard leaderboards.
4. **Convergence guarantees:** The cantilever generation loop converges empirically but lacks formal convergence proofs.
5. **Sparse chord coverage:** With only 5 bits set per byte, the 257-bit chord is very sparse. The capacity limits of this representation under heavy superposition need formal characterization.

10.7 Evaluation Roadmap

Planned Ablation Studies.

- Disable EigenMode steering (flat popcount only) to isolate the contribution of toroidal phase alignment.
- Vary chord width ($D \in \{128, 257, 512, 1024\}$) to characterize the sparsity–capacity trade-off.
- Vary manifold depth (Cubic-only vs. Icosahedral) to measure the entropy routing contribution.
- Disable winding counter to test syntactic depth discrimination.

Scaling Experiments.

- Dictionary size vs. BestFill latency: characterize the empirical scaling constant for 1K to 10M manifolds.
- Chord width vs. retrieval accuracy: determine the minimum D for each modality.
- Compression ratio vs. corpus size: validate the predicted increasing-compression-with-scale behavior from Zipfian theory (Section 0.31).

Novel Modality: Protein Secondary Structure. As a test of the architecture's genuine modality-agnosticism, we plan experiments on protein secondary structure prediction using amino acid sequences as raw byte input. This domain provides a particularly compelling test case:

1. The input alphabet (20 amino acid letters) is a strict subset of the byte range, requiring no architectural modification.
2. Protein backbone geometry is governed by torsion angle pairs (ϕ, ψ) that naturally inhabit a torus $S^1 \times S^1$— the same manifold topology that EigenMode discovers from co-occurrence statistics.
3. Secondary structure elements (α-helices, β-sheets) are defined by *periodic local patterns* in residue sequence—precisely the type of non-commutative rotational structure that the manifold is designed to encode.
4. The $O(1)$ rotation and popcount-based search offer an asymptotic advantage over molecular dynamics ($O(N^2)$ per timestep) for conformational space exploration.

We presented a physics-native cognitive architecture that frames sequence generation as a boundary value problem over discrete bitwise geometry. The system rests on four concrete architectural pillars:

1. A **257-bit Chord** representation with deterministic prime-basis encoding, where bitwise operations execute discrete wave interference at single-cycle throughput.
2. A **Nested SO(3) Fractal Manifold** that encodes syntactic sequence through non-commutative A_5 rotations, with $O(1)$ virtual mitosis for entropy pressure relief.
3. A **5-pass GPU BestFill pipeline** that hierarchically eliminates candidates using integer winding/state filters, SIMD popcount, and precomputed geodesic disambiguation—scanning millions of manifolds in sub-millisecond latency.
4. A **Toroidal Eigenmode** navigation layer that steers generation along dense trajectories on an $S^1 \times S^1$ phase torus, preventing echo-loop entrapment and enabling topological closure detection.

Across the reported experiments, the system demonstrates gradient-free inference, modality-agnostic byte-level processing, and hardware-native parallel retrieval. The identified failure modes—insufficient energy selectivity and shallow propagation depth—converge on a single architectural surface: the interaction between popcount gating and manifold hierarchy depth.

Hyperdimensional Computing and VSAs. The Chord representation is most directly situated within Hyperdimensional Computing (HDC) and Vector Symbolic Architectures (VSA). Classical VSA frameworks use high-dimensional random vectors ($D \geq 10{,}000$) with binding, bundling, and permutation operations. Our 257-bit Chord provides a far more aggressive compression, trading dimensionality for the non-commutative rotational manifold that escapes the Shannon capacity limit via topological entropy routing (Section 2).

Discrete Symmetry Groups in ML. Equivariant neural networks exploit continuous symmetry groups (rotations, translations) to build inductive biases. Our use of the discrete alternating group A_5 (the symmetry group of the icosahedron, order 60) is architecturally distinct: rather than learning equivariant features, we use A_5 permutations as the *computational substrate itself*—syntactic events directly manipulate geometric state through group actions.

Content-Addressable Memory. Hopfield networks, modern Hopfield networks, and transformer-based associative memories store patterns as energy minima or attention-weighted retrievals. Our system differs in substrate: memory is not a learned weight matrix but a persistent geometric index of bitwise manifolds. New observations reshape memory immediately via deterministic insertion.

Byte-Level Models. ByT5 and MEGABYTE process raw bytes to eliminate vocabulary failures, but retain learned encoders and autoregressive decoding. Our Universal Tokenizer assigns deterministic identities without learning, and downstream inference is parallel BestFill rather than autoregressive.

Non-Autoregressive Generation. Masked language models and diffusion-based text generators produce tokens in parallel but struggle with output coherence. Our BVP formulation resolves inter-position dependencies through shared manifold topology—structural coherence is a geometric constraint, not a decoding heuristic.